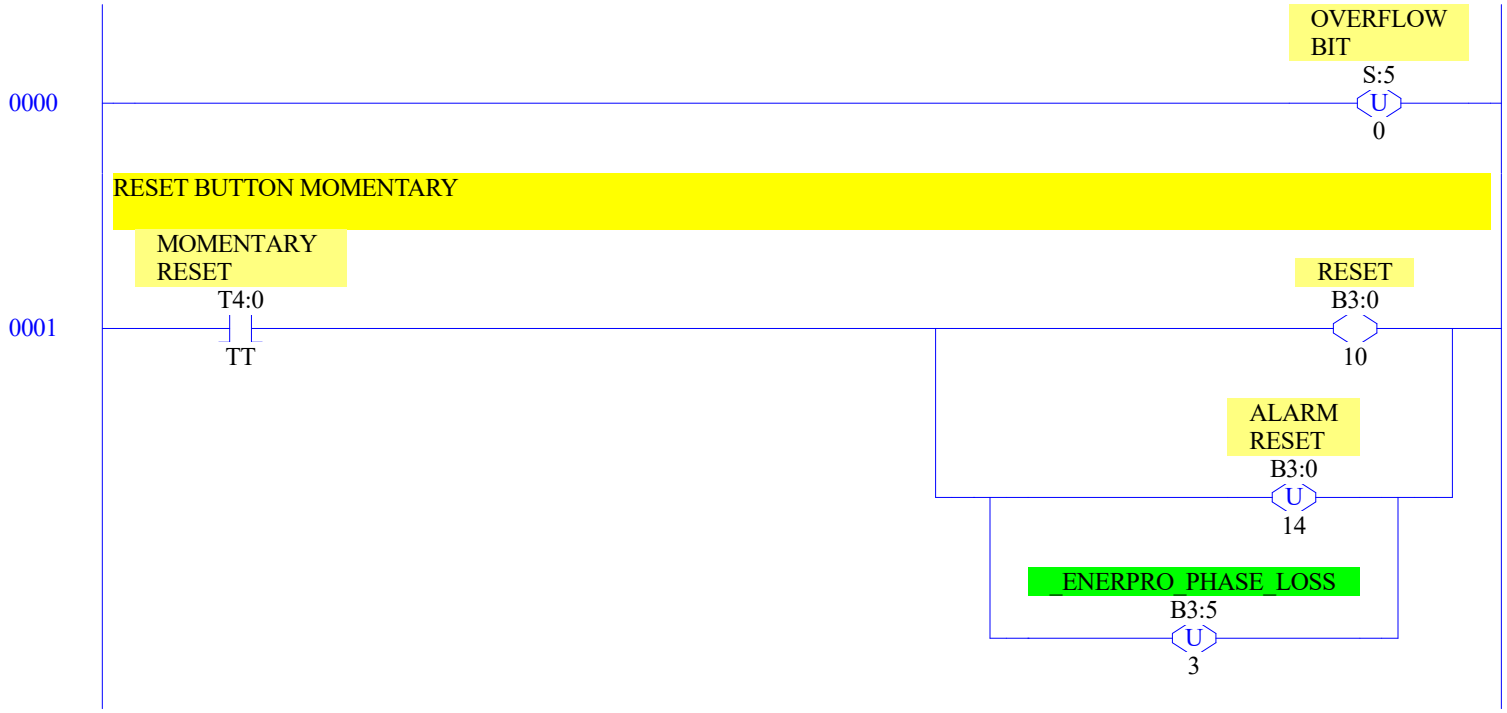
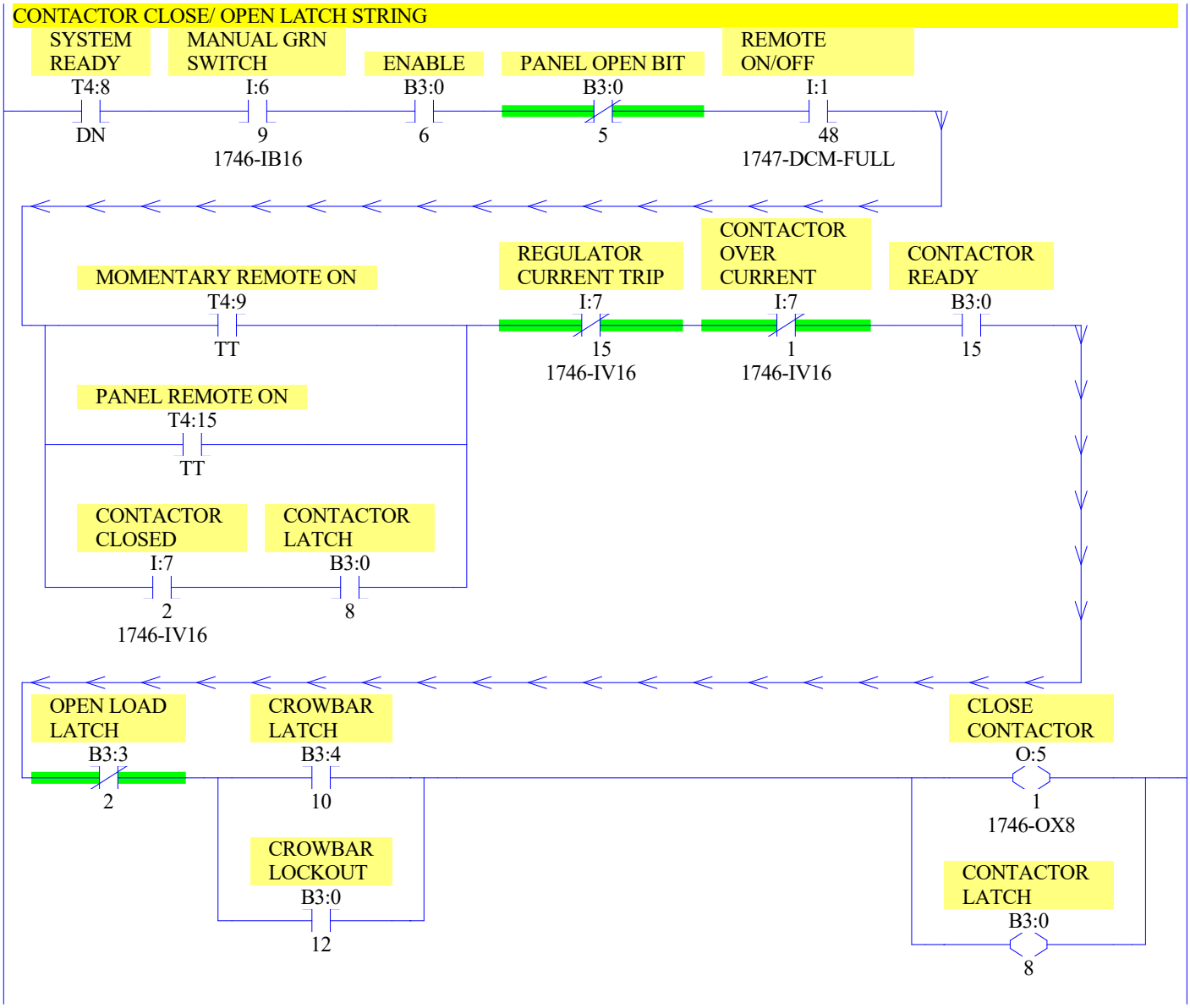
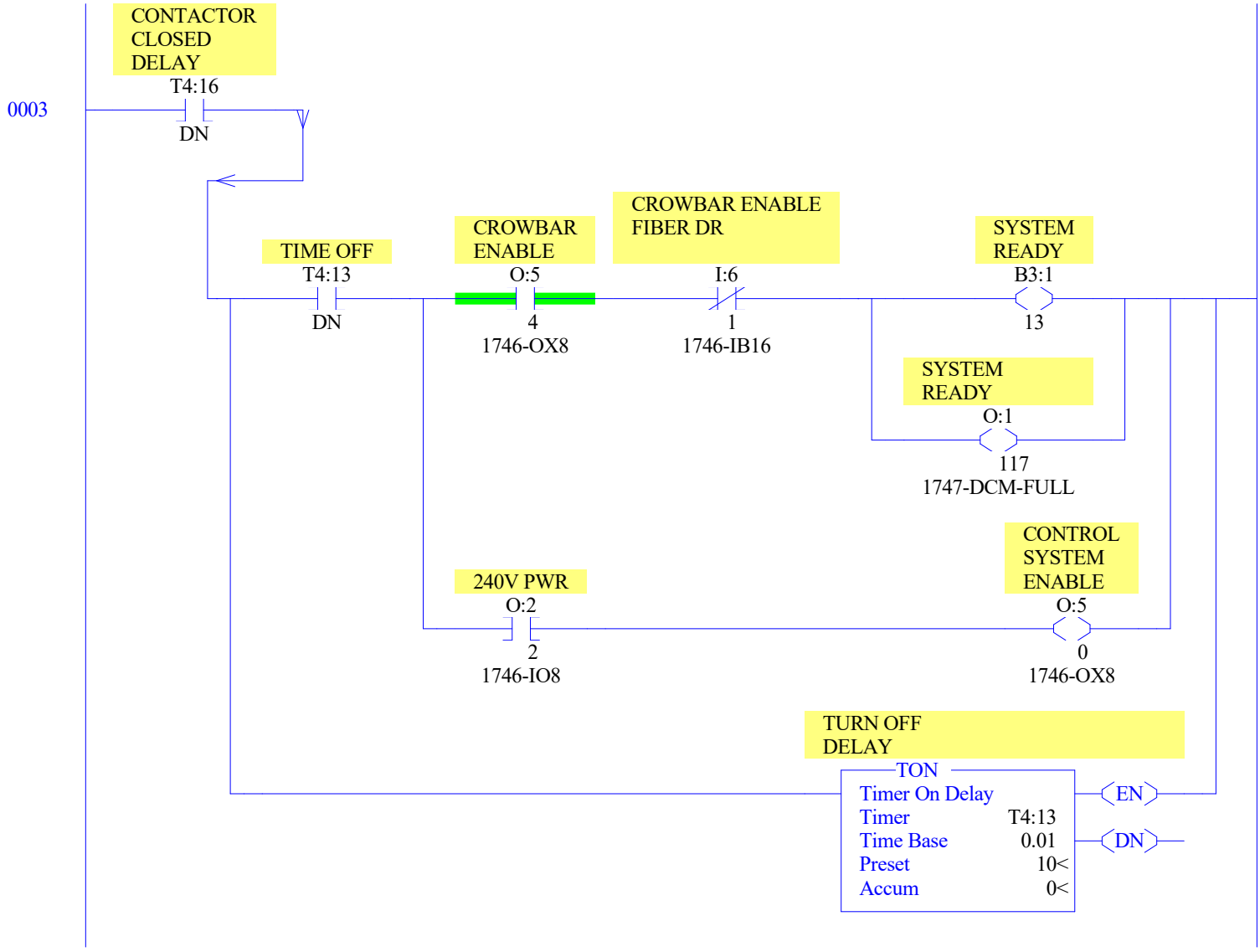
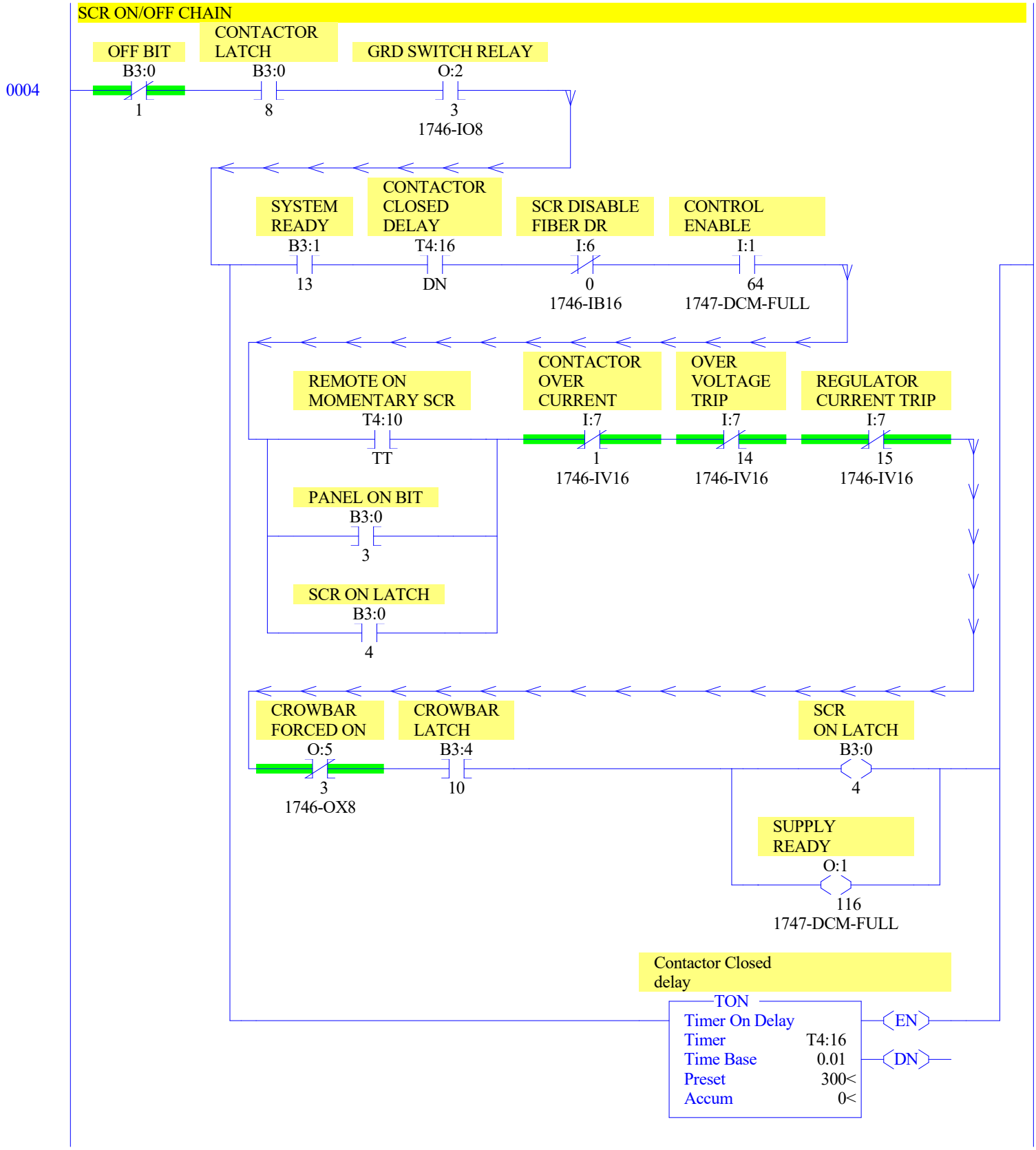
Program File List

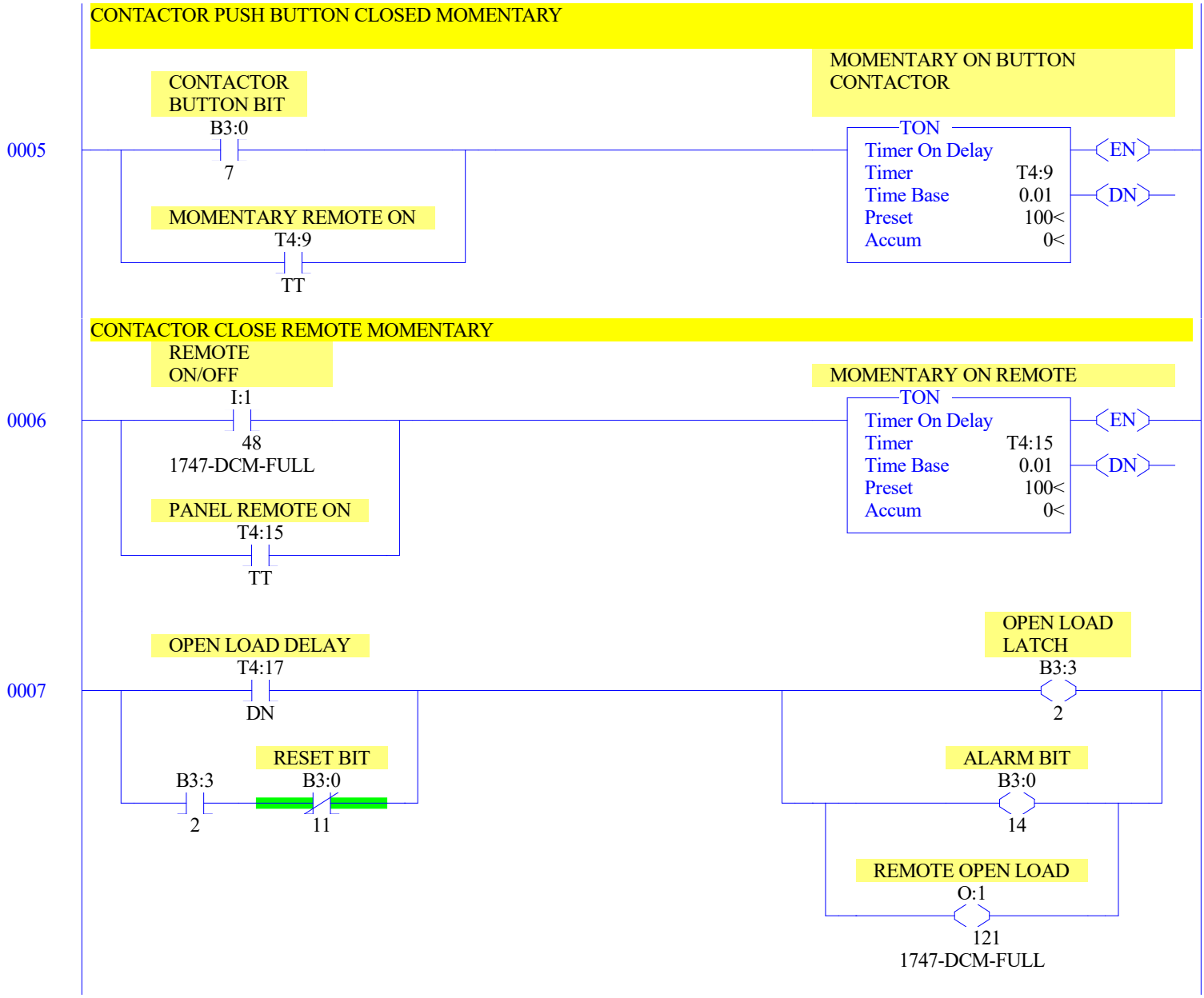
Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
	2	LADDER	120	No	5557
COPY	3	LADDER	6	No	108
SCALE	4	LADDER	5	No	159


0000
OVERFLOW BIT
S:5
U
0
RESET BUTTON MOMENTARY
0001
MOMENTARY RESET
T4:0
TT
RESET
B3:0
10
ALARM RESET
B3:0
U
14
_ENERPRO_PHASE_LOSS
B3:5
U
3


CONTACTOR CLOSE/ OPEN LATCH STRING
0002
SYSTEM READY
T4:8
DN
MANUAL GRN SWITCH
I:6
9
1746-IB16
ENABLE
B3:0
6
PANEL OPEN BIT
B3:0
5
REMOTE ON/OFF
I:1
48
1747-DCM-FULL
MOMENTARY REMOTE ON
T4:9
TT
REGULATOR CURRENT TRIP
I:7
15
1746-IV16
CONTACTOR OVER CURRENT
I:7
1
1746-IV16
CONTACTOR READY
B3:0
15
PANEL REMOTE ON
T4:15
TT
CONTACTOR CLOSED
I:7
2
1746-IV16
CONTACTOR LATCH
B3:0
8
OPEN LOAD LATCH
B3:3
2
CROWBAR LATCH
B3:4
10
CROWBAR LOCKOUT
B3:0
12
CLOSE CONTACTOR
O:5
1
1746-OX8
CONTACTOR LATCH
B3:0
8


0003
CONTACTOR CLOSED DELAY
T4:16
DN
TIME OFF
T4:13
DN
CROWBAR ENABLE
O:5
4
1746-OX8
CROWBAR ENABLE FIBER DR
I:6
1
1746-IB16
SYSTEM READY
B3:1
13
SYSTEM READY
O:1
117
1747-DCM-FULL
240V PWR
O:2
2
1746-IO8
CONTROL SYSTEM ENABLE
O:5
0
1746-OX8
TURN OFF DELAY
TON
Timer On Delay
Timer T4:13
Time Base 0.01
Preset 10<
Accum 0<
EN
DN


SCR ON/OFF CHAIN
0004
OFF BIT
B3:0
1
CONTACTOR LATCH
B3:0
8
GRD SWITCH RELAY
O:2
3
1746-IO8
SYSTEM READY
B3:1
13
CONTACTOR CLOSED DELAY
T4:16
DN
SCR DISABLE FIBER DR
I:6
0
1746-IB16
CONTROL ENABLE
I:1
64
1747-DCM-FULL
REMOTE ON MOMENTARY SCR
T4:10
TT
CONTACTOR OVER CURRENT
I:7
1
1746-IV16
OVER VOLTAGE TRIP
I:7
14
1746-IV16
REGULATOR CURRENT TRIP
I:7
15
1746-IV16
PANEL ON BIT
B3:0
3
SCR ON LATCH
B3:0
4
CROWBAR FORCED ON
O:5
3
1746-OX8
CROWBAR LATCH
B3:4
10
SCR ON LATCH
B3:0
4
SUPPLY READY
O:1
116
1747-DCM-FULL
Contactor Closed delay
TON
Timer On Delay
Timer T4:16
Time Base 0.01
Preset 300<
Accum 0<
EN
DN


CONTACTOR PUSH BUTTON CLOSED MOMENTARY
0005
CONTACTOR BUTTON BIT
B3:0
7
MOMENTARY REMOTE ON
T4:9
TT
MOMENTARY ON BUTTON CONTACTOR
TON
Timer On Delay
Timer T4:9
Time Base 0.01
Preset 100<
Accum 0<
EN
DN
CONTACTOR CLOSE REMOTE MOMENTARY
0006
REMOTE ON/OFF
I:1
48
1747-DCM-FULL
PANEL REMOTE ON
T4:15
TT
MOMENTARY ON REMOTE
TON
Timer On Delay
Timer T4:15
Time Base 0.01
Preset 100<
Accum 0<
EN
DN
0007
OPEN LOAD DELAY
T4:17
DN
B3:3
2
RESET BIT
B3:0
11
OPEN LOAD LATCH
B3:3
2
ALARM BIT
B3:0
14
REMOTE OPEN LOAD
O:1
121
1747-DCM-FULL

OPEN LOAD DETECTOR IF VOLTAGE > N7:76,78 AND CURRENT LESS THAN N7:77,79

0008

DC CURRENT MONITOR

LES
Less Than (A<B)
Source A N7:17 11<
Source B N7:77 10<

VOLTAGE MONITOR #1

GRT
Greater Than (A>B)
Source A N7:15 1<
Source B N7:76 5000<

AC CURRENT MONITOR

LES
Less Than (A<B)
Source A N7:14 0<
Source B N7:79 30<

VOLTAGE MONITOR #2

GRT
Greater Than (A>B)
Source A N7:16 0<
Source B N7:78 5000<

OPEN LOAD DELAY

TON
Timer On Delay
Timer T4:17
Time Base 0.01
Preset 100<
Accum 0<
EN
DN

SCR ON/OFF CONTROLS MOMENTARY

0009

CONTROL ENABLE

I:1
64
1747-DCM-FULL

REMOTE MOMENTARY ON SCR'S

TON
Timer On Delay
Timer T4:10
Time Base 0.01
Preset 50<
Accum 0<
EN
DN

TURN OFF SEQUENCE

0010

SCR ON LATCH
B3:0
4

ENABLE DELAY
TON
Timer On Delay
Timer T4:5
Time Base 0.01
Preset 300<
Accum 0<
EN
DN

FAST INHIBIT
B3:0
L
13

ENABLE DELAY
T4:5
DN

REGULATOR ON
B3:0
2

REFERANCE VOLTAGE OR MAXIMUM

0011

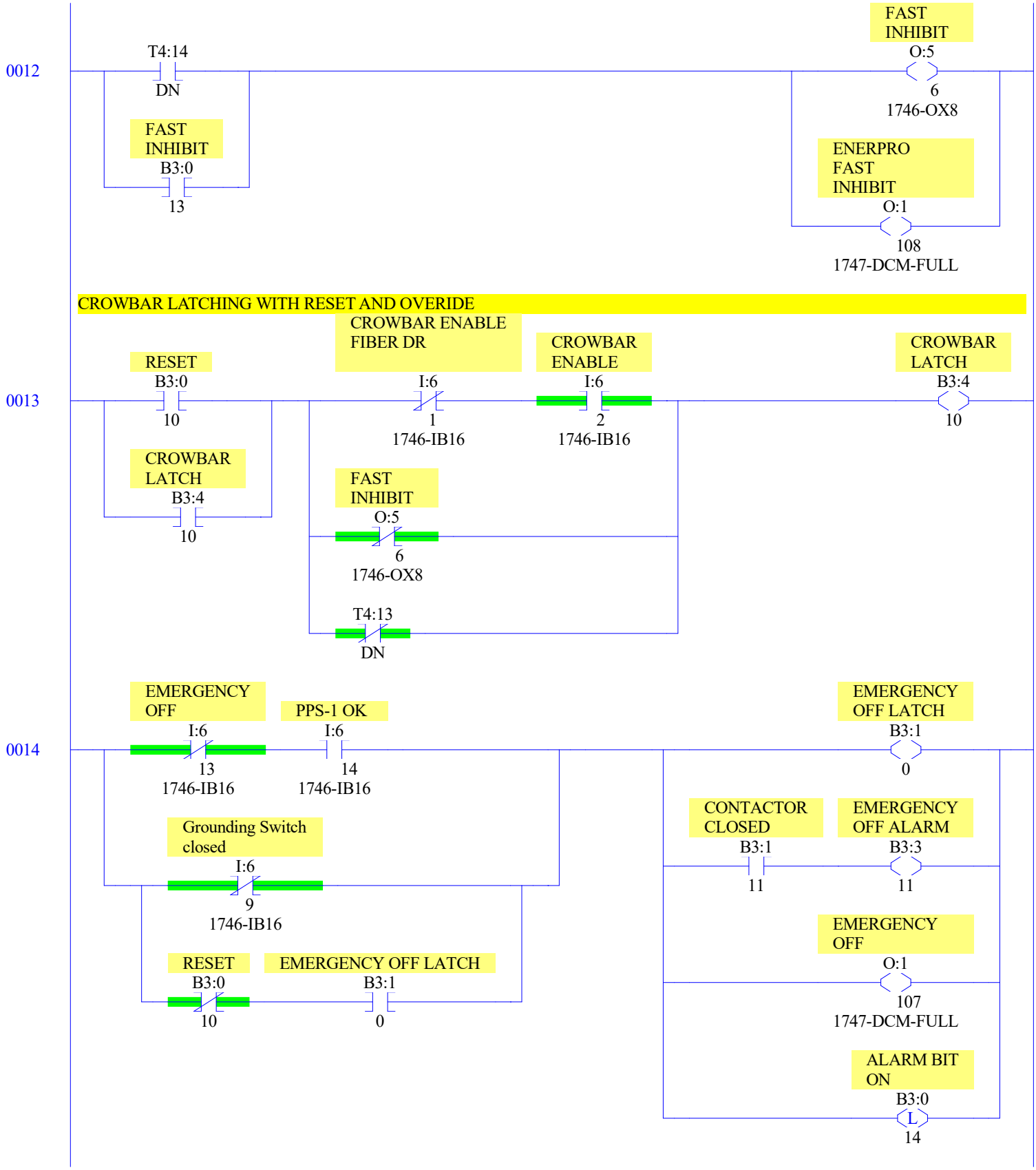
REGULATOR ON
B3:0
2

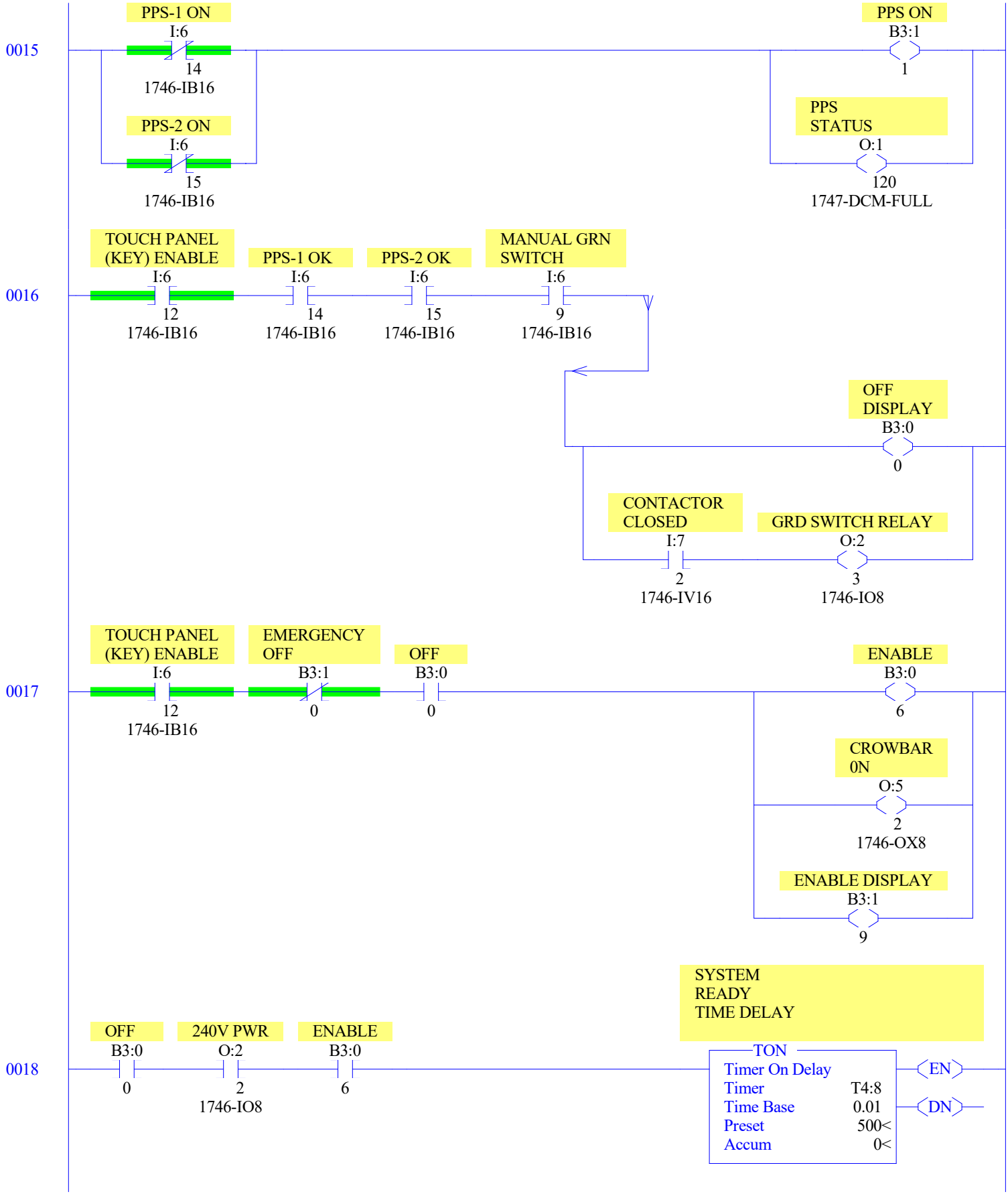
REFERANCE CONTROL
MOV
Move
Source 0
0<
Dest N7:10
0<

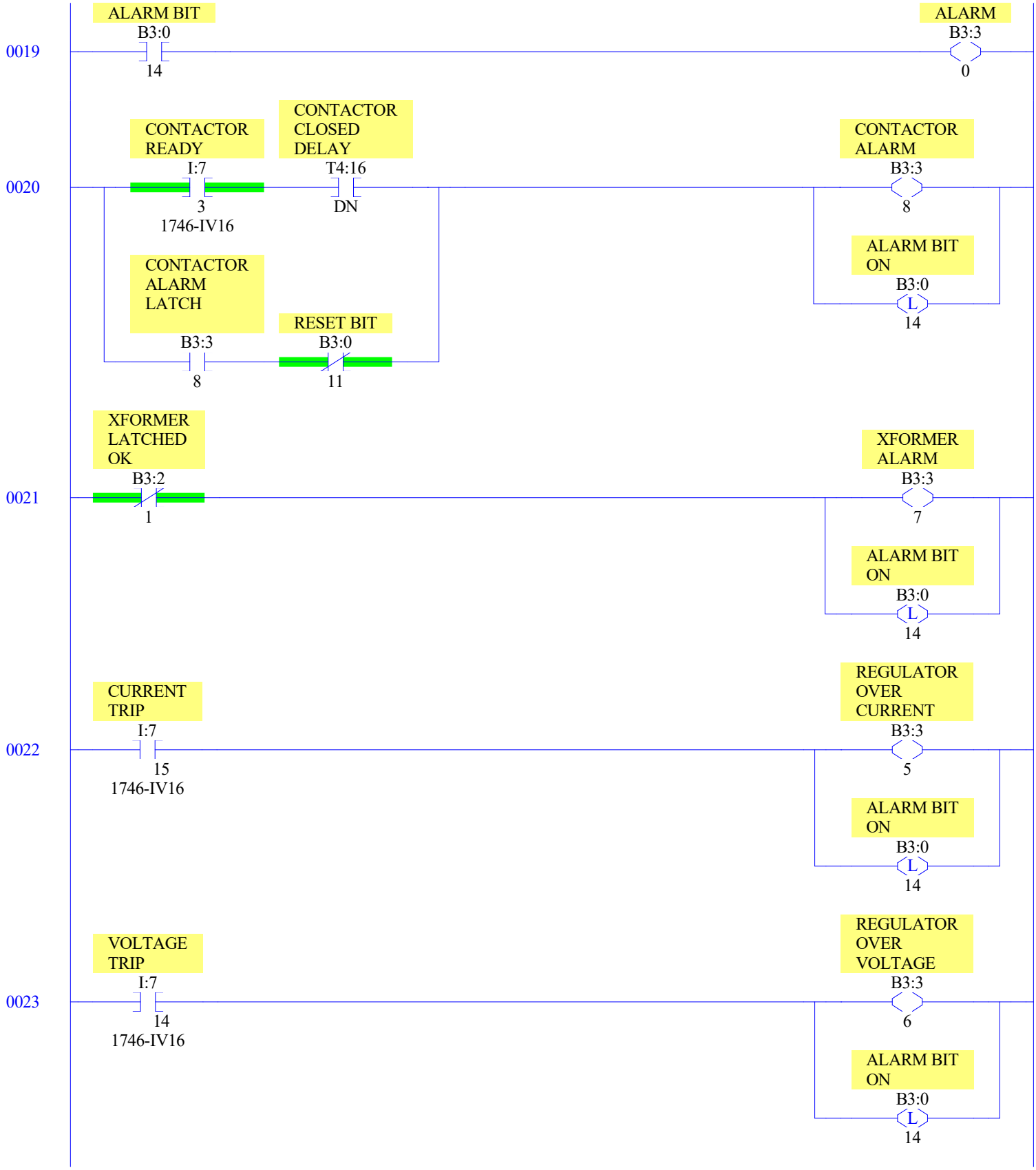
PHASE ANGLE CONTROL
MOV
Move
Source 0
0<
Dest N7:11
0<

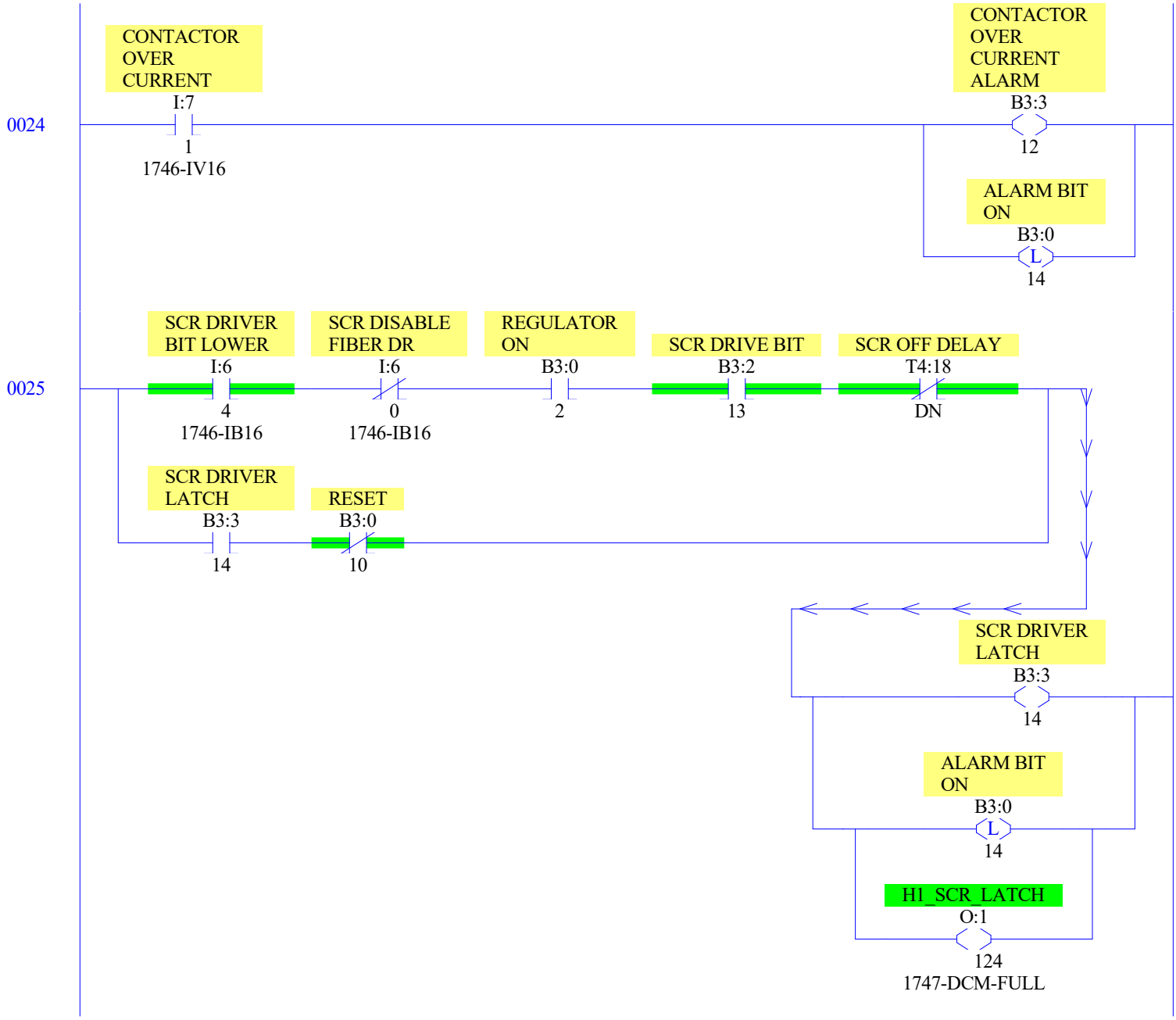
T4:14
DN

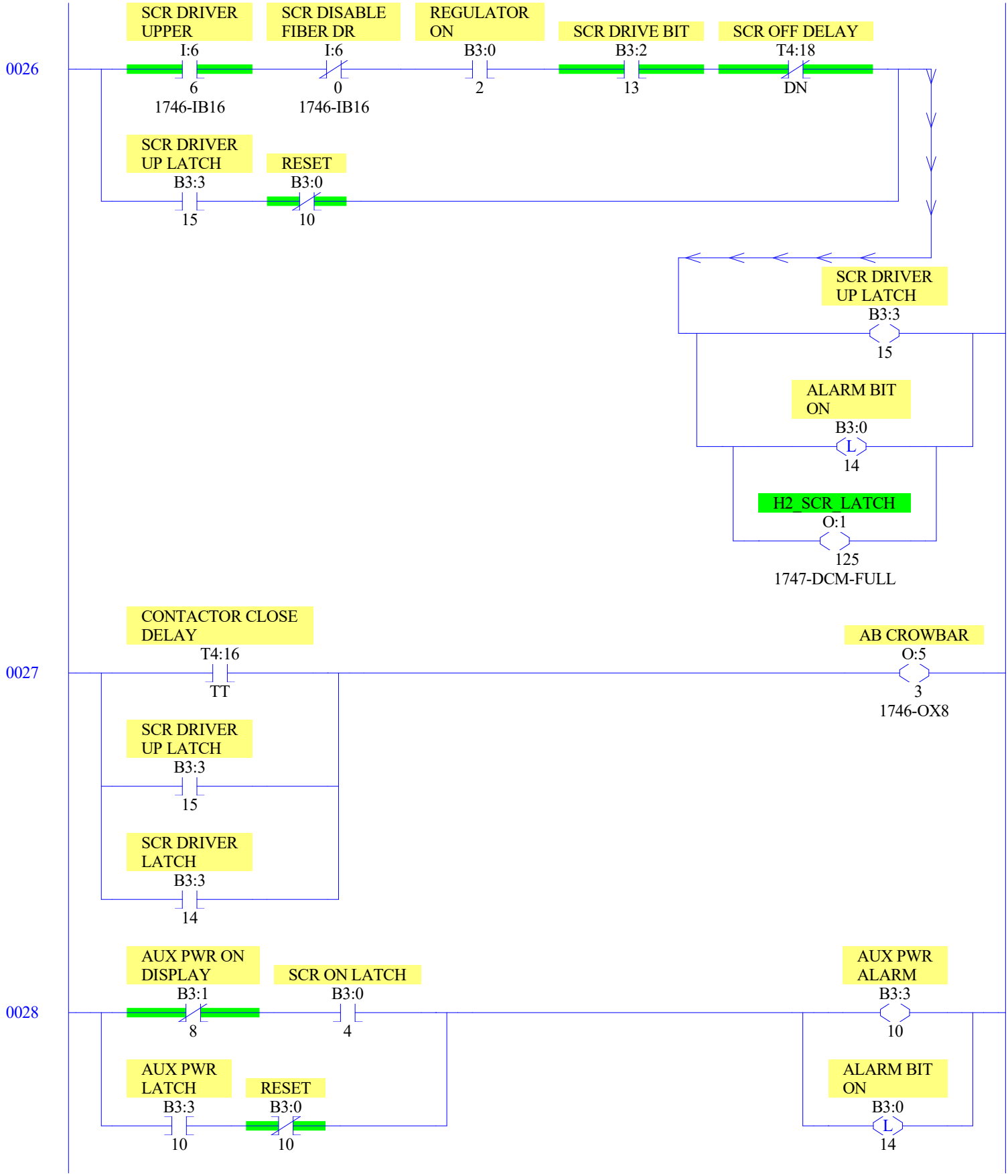
FAST INHIBIT
B3:0
U
13

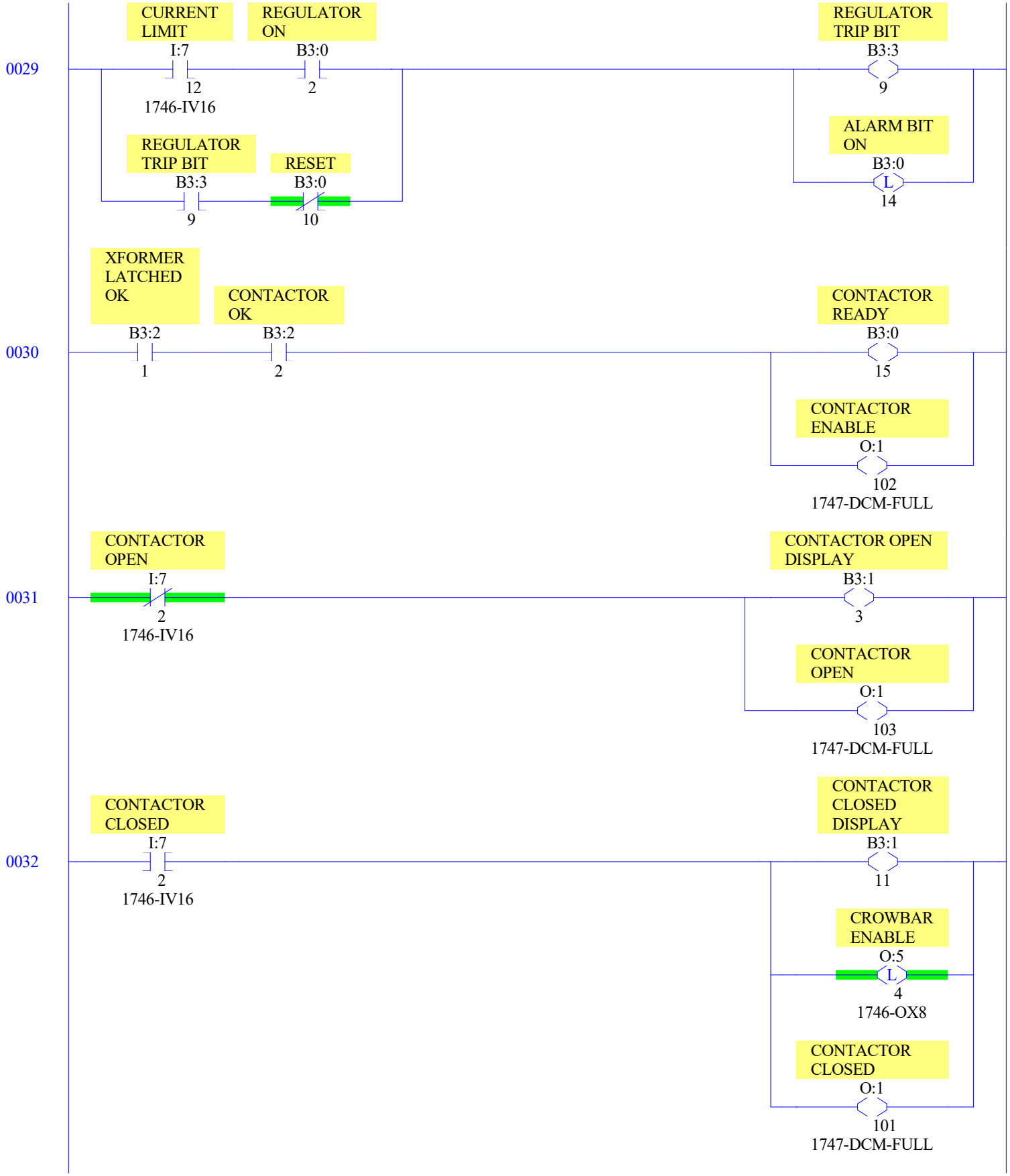

0012
T4:14
DN
FAST INHIBIT
B3:0
13
FAST INHIBIT
O:5
6
1746-OX8
ENERPRO FAST INHIBIT
O:1
108
1747-DCM-FULL
CROWBAR LATCHING WITH RESET AND OVERIDE
0013
RESET
B3:0
10
CROWBAR LATCH
B3:4
10
CROWBAR ENABLE FIBER DR
I:6
1
1746-IB16
CROWBAR ENABLE
I:6
2
1746-IB16
FAST INHIBIT
O:5
6
1746-OX8
T4:13
DN
CROWBAR LATCH
B3:4
10
0014
EMERGENCY OFF
I:6
13
1746-IB16
PPS-1 OK
I:6
14
1746-IB16
Grounding Switch closed
I:6
9
1746-IB16
RESET
B3:0
10
EMERGENCY OFF LATCH
B3:1
0
EMERGENCY OFF LATCH
B3:1
0
CONTACTOR CLOSED
B3:1
11
EMERGENCY OFF ALARM
B3:3
11
EMERGENCY OFF
O:1
107
1747-DCM-FULL
ALARM BIT ON
B3:0
L
14

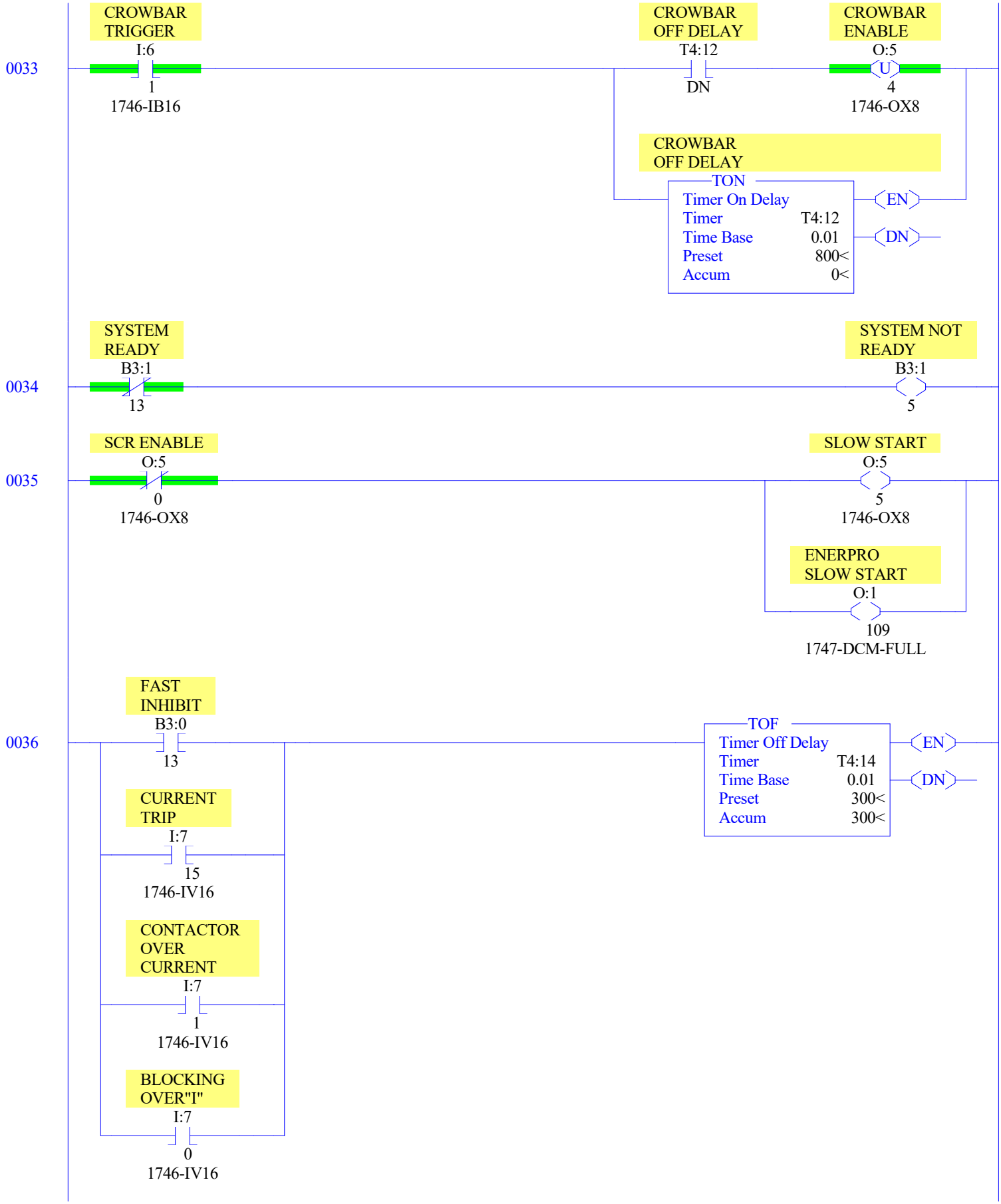

0015
PPS-1 ON
I:6
14
1746-IB16
PPS-2 ON
I:6
15
1746-IB16
PPS ON
B3:1
1
PPS STATUS
O:1
120
1747-DCM-FULL
0016
TOUCH PANEL (KEY) ENABLE
I:6
12
1746-IB16
PPS-1 OK
I:6
14
1746-IB16
PPS-2 OK
I:6
15
1746-IB16
MANUAL GRN SWITCH
I:6
9
1746-IB16
OFF DISPLAY
B3:0
0
CONTACTOR CLOSED
I:7
2
1746-IV16
GRD SWITCH RELAY
O:2
3
1746-IO8
0017
TOUCH PANEL (KEY) ENABLE
I:6
12
1746-IB16
EMERGENCY OFF
B3:1
0
OFF
B3:0
0
ENABLE
B3:0
6
CROWBAR 0N
O:5
2
1746-OX8
ENABLE DISPLAY
B3:1
9
SYSTEM READY TIME DELAY
0018
OFF
B3:0
0
240V PWR
O:2
2
1746-IO8
ENABLE
B3:0
6
TON
Timer On Delay
Timer T4:8
Time Base 0.01
Preset 500<
Accum 0<
EN
DN

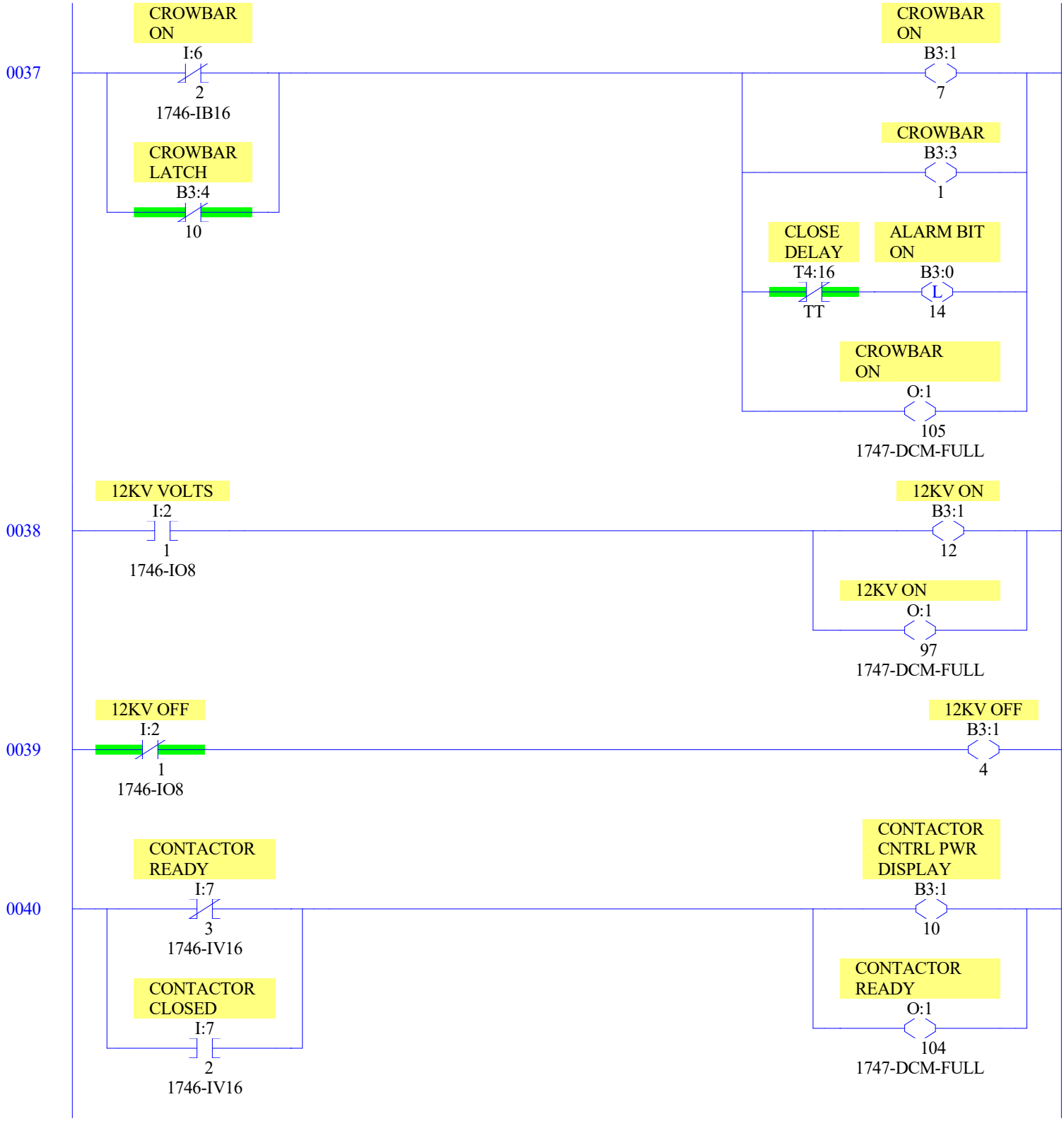

0019
ALARM BIT
B3:0
14
ALARM
B3:3
0
0020
CONTACTOR READY
I:7
3
1746-IV16
CONTACTOR CLOSED DELAY
T4:16
DN
CONTACTOR ALARM LATCH
B3:3
8
RESET BIT
B3:0
11
CONTACTOR ALARM
B3:3
8
ALARM BIT ON
B3:0
L
14
0021
XFORMER LATCHED OK
B3:2
1
XFORMER ALARM
B3:3
7
ALARM BIT ON
B3:0
L
14
0022
CURRENT TRIP
I:7
15
1746-IV16
REGULATOR OVER CURRENT
B3:3
5
ALARM BIT ON
B3:0
L
14
0023
VOLTAGE TRIP
I:7
14
1746-IV16
REGULATOR OVER VOLTAGE
B3:3
6
ALARM BIT ON
B3:0
L
14

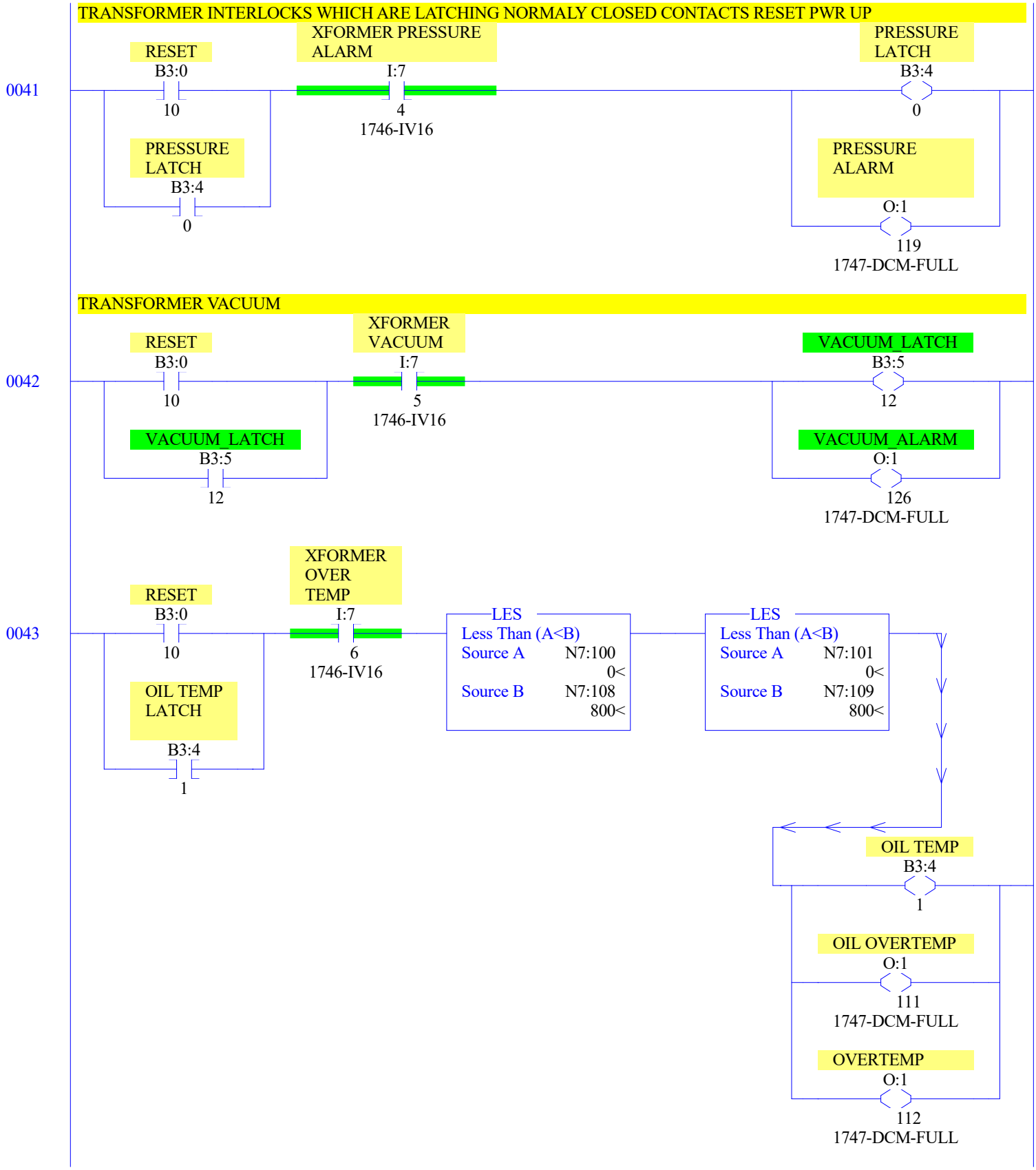

0024
CONTACTOR OVER CURRENT
I:7
1
1746-IV16
CONTACTOR OVER CURRENT ALARM
B3:3
12
ALARM BIT ON
B3:0
L
14
0025
SCR DRIVER BIT LOWER
I:6
4
1746-IB16
SCR DISABLE FIBER DR
I:6
0
1746-IB16
REGULATOR ON
B3:0
2
SCR DRIVE BIT
B3:2
13
SCR OFF DELAY
T4:18
DN
SCR DRIVER LATCH
B3:3
14
RESET
B3:0
10
SCR DRIVER LATCH
B3:3
14
ALARM BIT ON
B3:0
L
14
H1_SCR_LATCH
O:1
124
1747-DCM-FULL

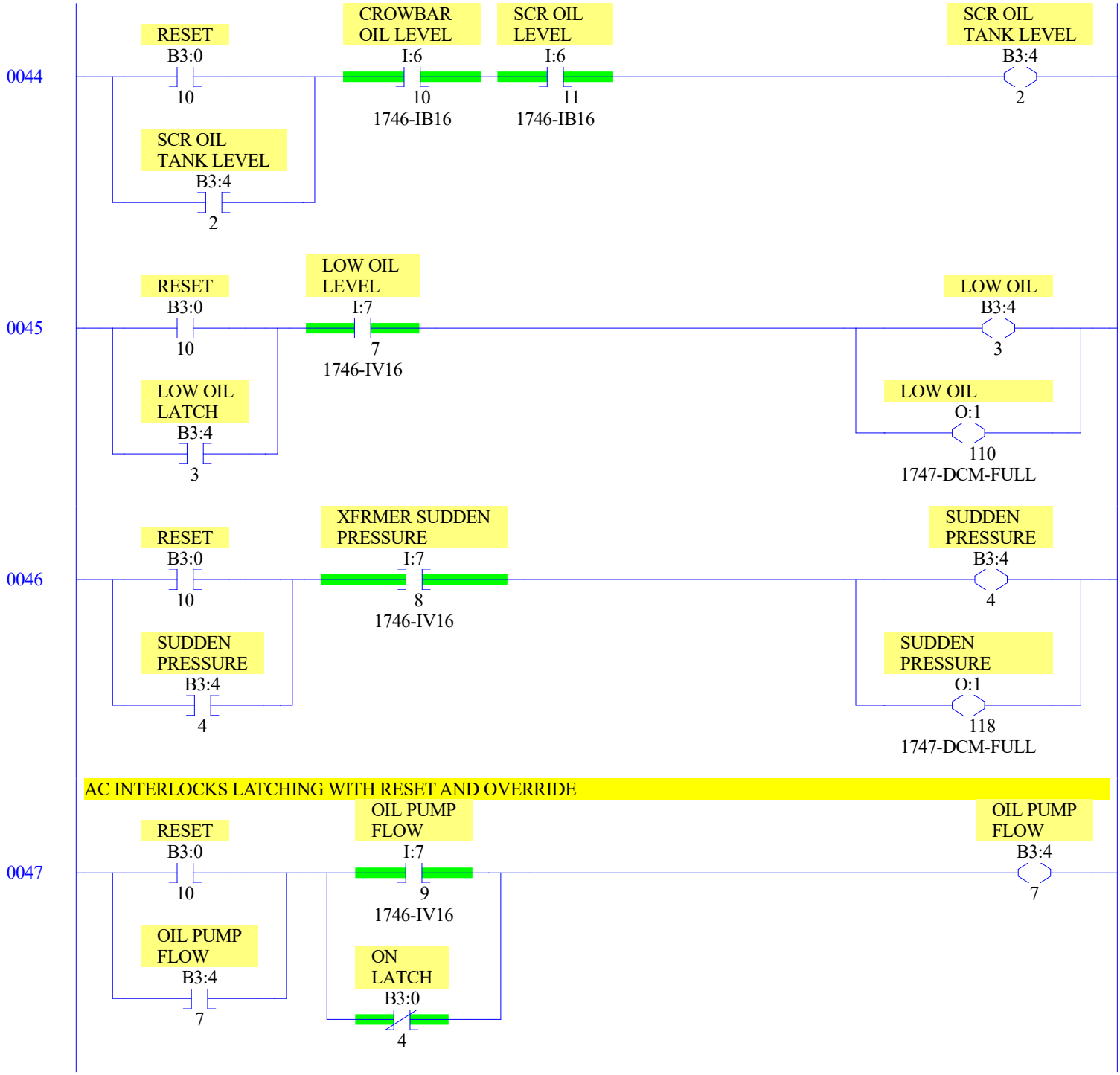

0026
SCR DRIVER UPPER
I:6
6
1746-IB16
SCR DISABLE FIBER DR
I:6
0
1746-IB16
REGULATOR ON
B3:0
2
SCR DRIVE BIT
B3:2
13
SCR OFF DELAY
T4:18
DN
SCR DRIVER UP LATCH
B3:3
15
RESET
B3:0
10
SCR DRIVER UP LATCH
B3:3
15
ALARM BIT ON
B3:0
L
14
H2_SCR_LATCH
O:1
125
1747-DCM-FULL
0027
CONTACTOR CLOSE DELAY
T4:16
TT
AB CROWBAR
O:5
3
1746-OX8
SCR DRIVER UP LATCH
B3:3
15
SCR DRIVER LATCH
B3:3
14
0028
AUX PWR ON DISPLAY
B3:1
8
SCR ON LATCH
B3:0
4
AUX PWR ALARM
B3:3
10
AUX PWR LATCH
B3:3
10
RESET
B3:0
10
ALARM BIT ON
B3:0
L
14

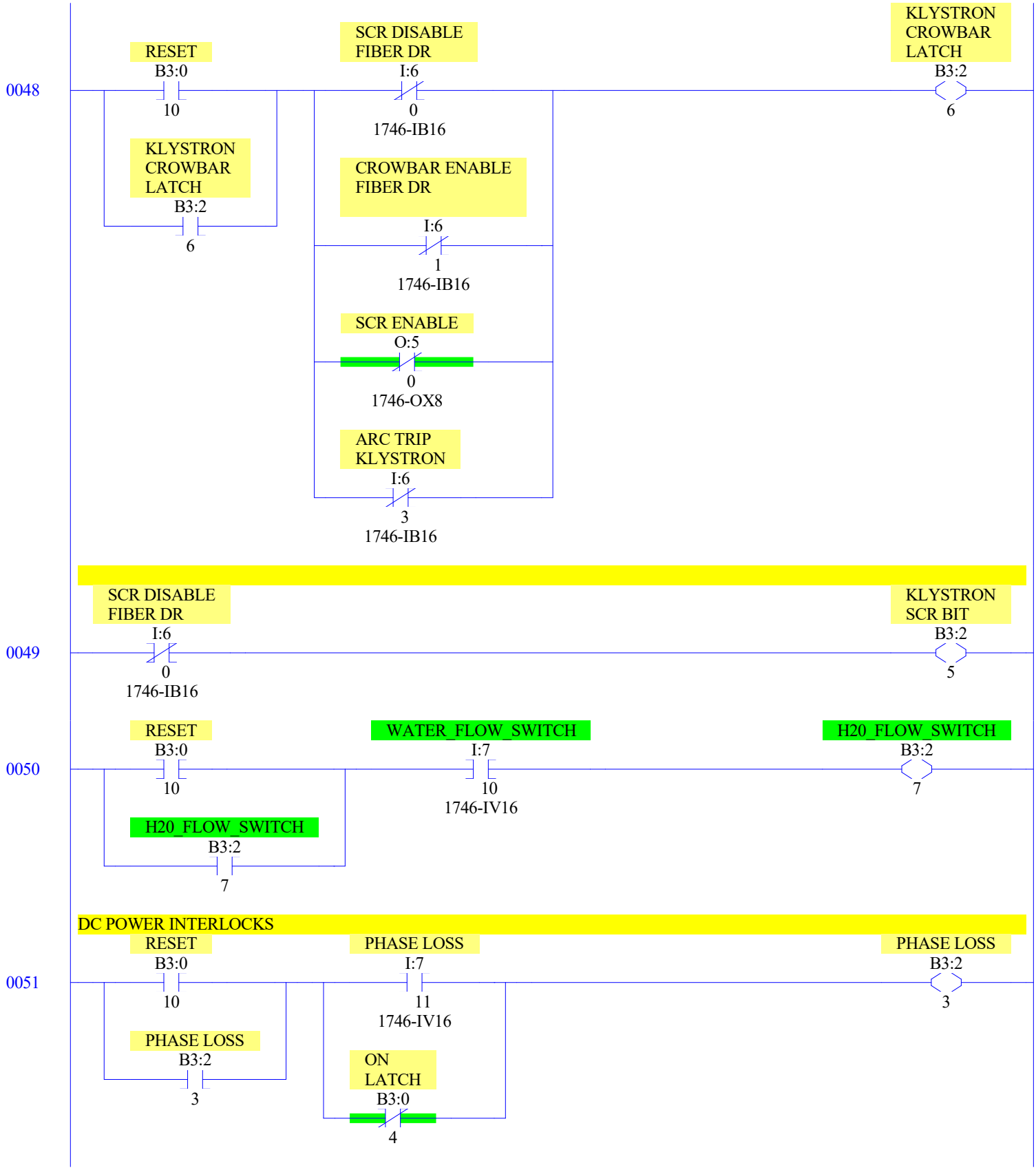

0029
CURRENT LIMIT
I:7
12
1746-IV16
REGULATOR ON
B3:0
2
REGULATOR TRIP BIT
B3:3
9
RESET
B3:0
10
REGULATOR TRIP BIT
B3:3
9
ALARM BIT ON
B3:0
L
14
0030
XFORMER LATCHED OK
B3:2
1
CONTACTOR OK
B3:2
2
CONTACTOR READY
B3:0
15
CONTACTOR ENABLE
O:1
102
1747-DCM-FULL
0031
CONTACTOR OPEN
I:7
2
1746-IV16
CONTACTOR OPEN DISPLAY
B3:1
3
CONTACTOR OPEN
O:1
103
1747-DCM-FULL
0032
CONTACTOR CLOSED
I:7
2
1746-IV16
CONTACTOR CLOSED DISPLAY
B3:1
11
CROWBAR ENABLE
O:5
L
4
1746-OX8
CONTACTOR CLOSED
O:1
101
1747-DCM-FULL

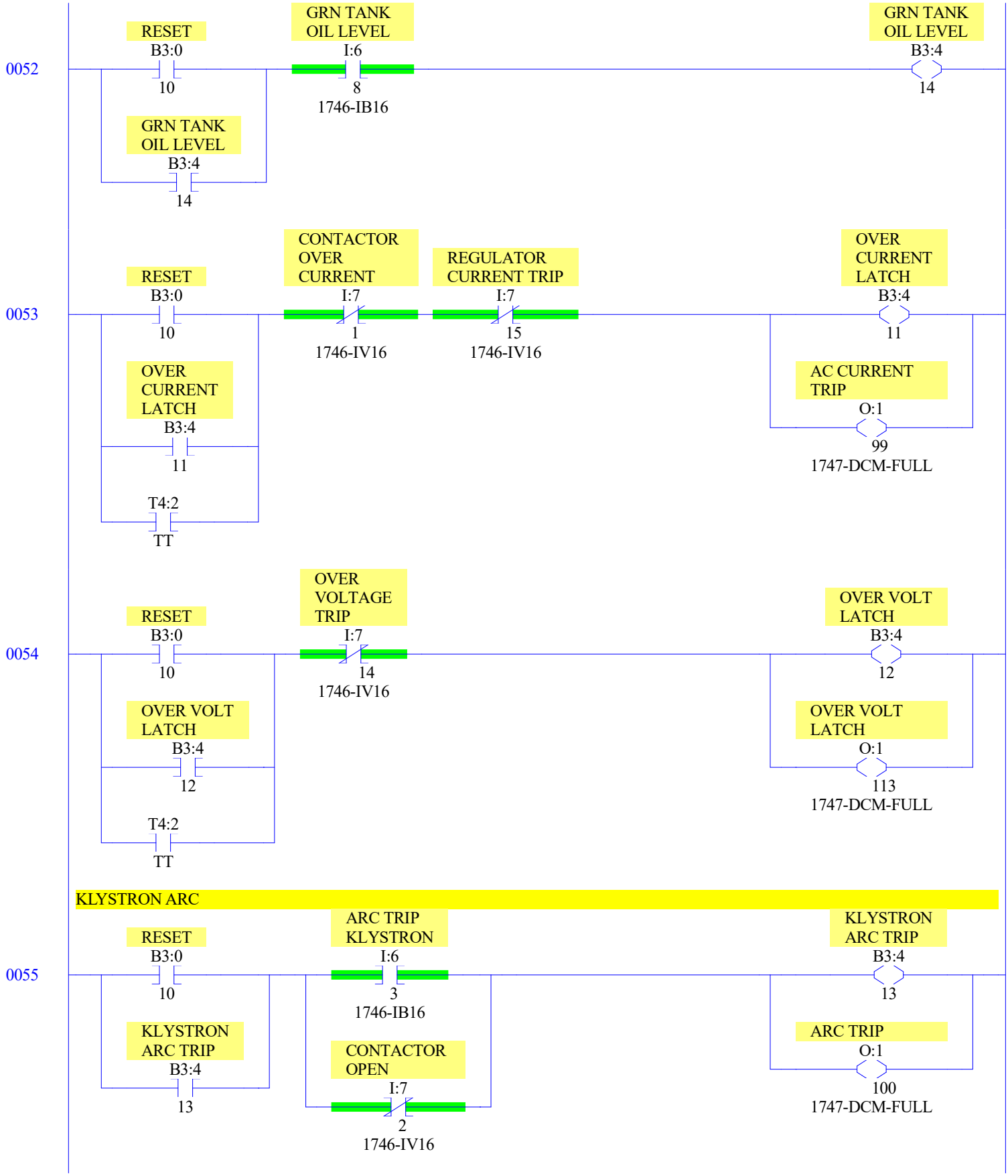

0033
CROWBAR TRIGGER
I:6
1
1746-IB16
CROWBAR OFF DELAY
T4:12
DN
CROWBAR ENABLE
O:5
U
4
1746-OX8
CROWBAR OFF DELAY
TON
Timer On Delay
Timer T4:12
Time Base 0.01
Preset 800<
Accum 0<
EN
DN
0034
SYSTEM READY
B3:1
13
SYSTEM NOT READY
B3:1
5
0035
SCR ENABLE
O:5
0
1746-OX8
SLOW START
O:5
5
1746-OX8
ENERPRO SLOW START
O:1
109
1747-DCM-FULL
0036
FAST INHIBIT
B3:0
13
CURRENT TRIP
I:7
15
1746-IV16
CONTACTOR OVER CURRENT
I:7
1
1746-IV16
BLOCKING OVER"I"
I:7
0
1746-IV16
TOF
Timer Off Delay
Timer T4:14
Time Base 0.01
Preset 300<
Accum 300<
EN
DN

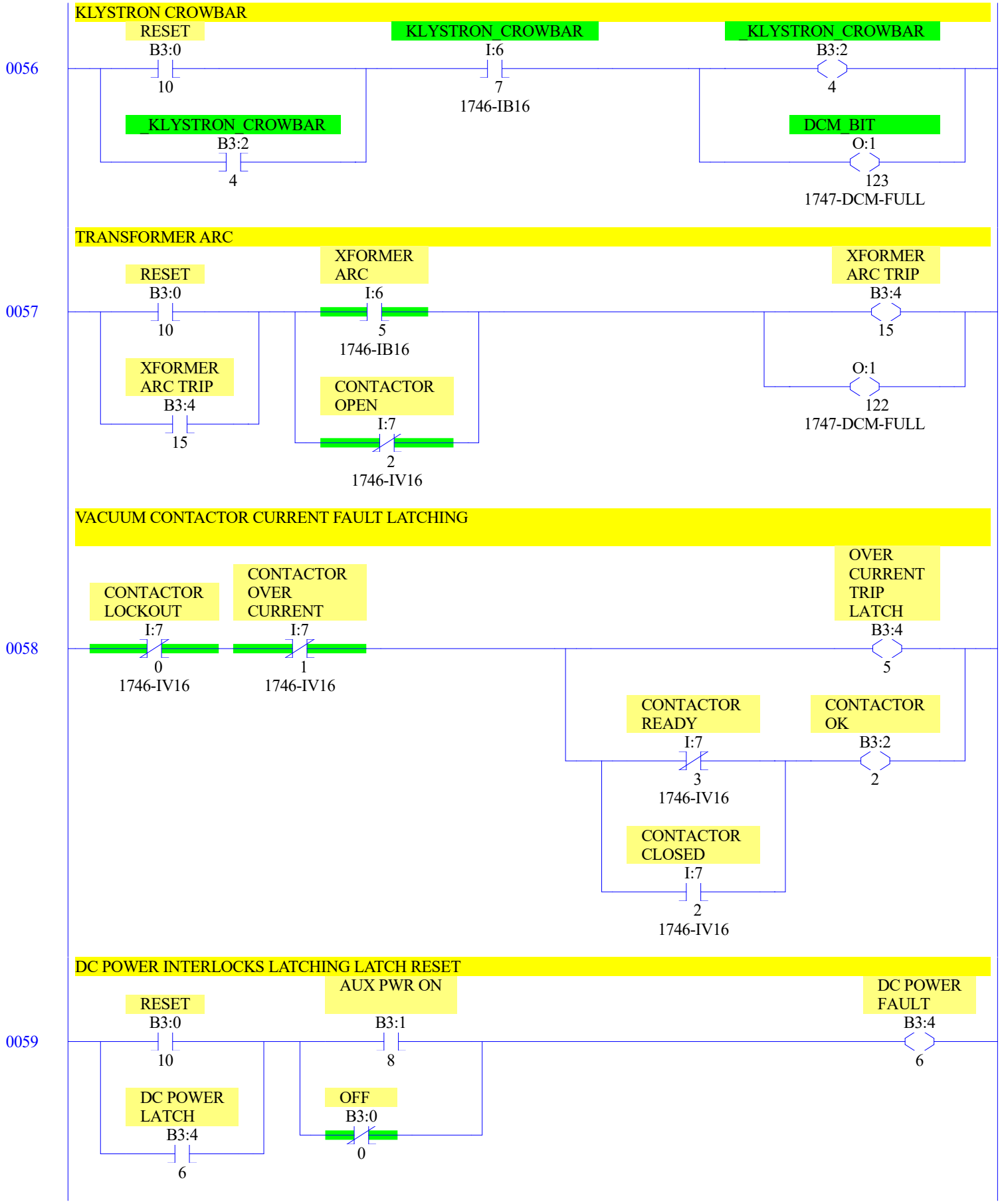

0037
CROWBAR ON
I:6
2
1746-IB16
CROWBAR LATCH
B3:4
10
CROWBAR ON
B3:1
7
CROWBAR
B3:3
1
CLOSE DELAY
T4:16
TT
ALARM BIT ON
B3:0
L
14
CROWBAR ON
O:1
105
1747-DCM-FULL
0038
12KV VOLTS
I:2
1
1746-IO8
12KV ON
B3:1
12
12KV ON
O:1
97
1747-DCM-FULL
0039
12KV OFF
I:2
1
1746-IO8
12KV OFF
B3:1
4
0040
CONTACTOR READY
I:7
3
1746-IV16
CONTACTOR CLOSED
I:7
2
1746-IV16
CONTACTOR CNTRL PWR DISPLAY
B3:1
10
CONTACTOR READY
O:1
104
1747-DCM-FULL

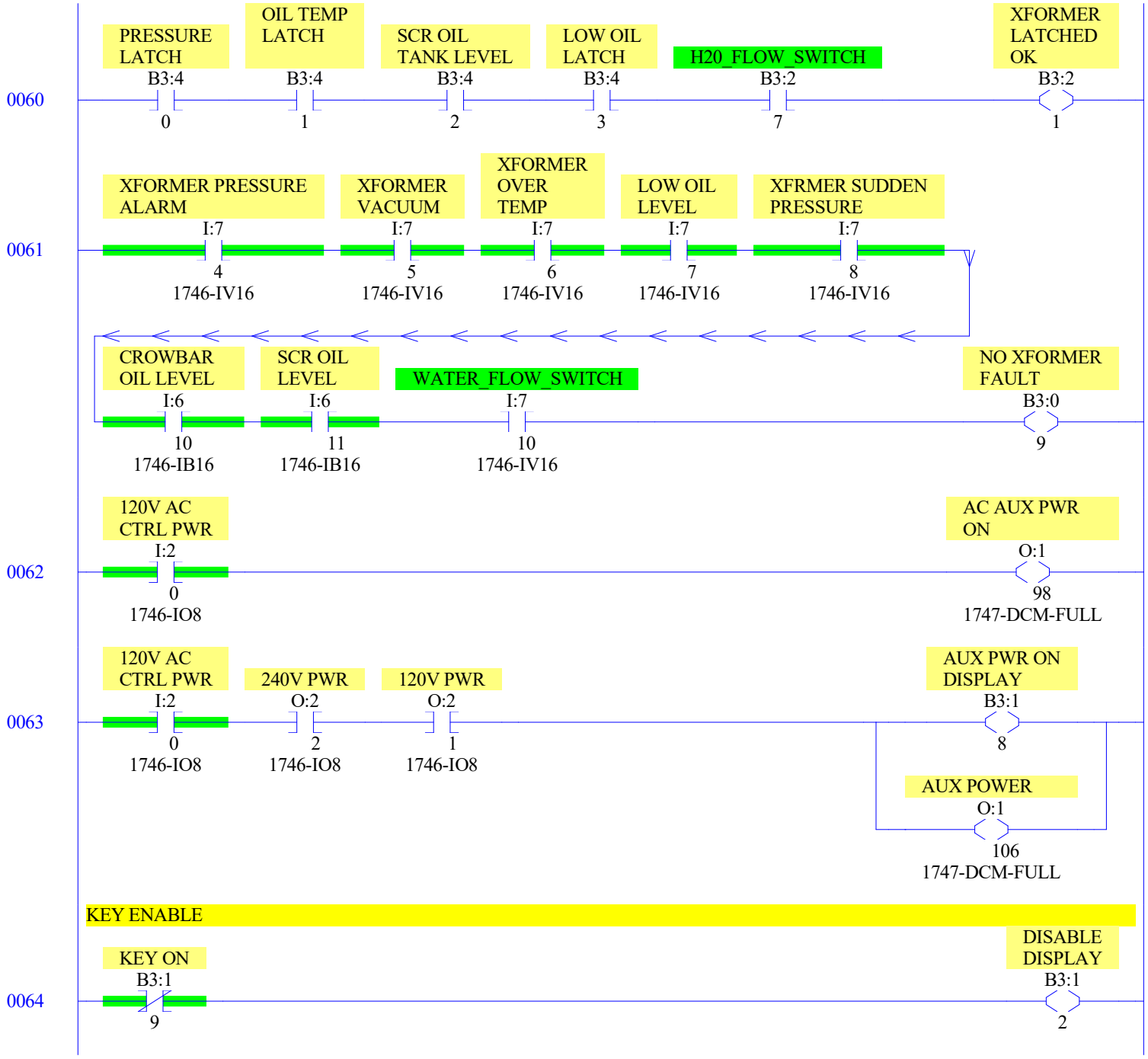

TRANSFORMER INTERLOCKS WHICH ARE LATCHING NORMALY CLOSED CONTACTS RESET PWR UP
0041
RESET
B3:0
10
PRESSURE LATCH
B3:4
0
XFORMER PRESSURE ALARM
I:7
4
1746-IV16
PRESSURE LATCH
B3:4
0
PRESSURE ALARM
O:1
119
1747-DCM-FULL
TRANSFORMER VACUUM
0042
RESET
B3:0
10
VACUUM_LATCH
B3:5
12
XFORMER VACUUM
I:7
5
1746-IV16
VACUUM_LATCH
B3:5
12
VACUUM_ALARM
O:1
126
1747-DCM-FULL
0043
RESET
B3:0
10
OIL TEMP LATCH
B3:4
1
XFORMER OVER TEMP
I:7
6
1746-IV16
LES
Less Than (A<B)
Source A N7:100
0<
Source B N7:108
800<
LES
Less Than (A<B)
Source A N7:101
0<
Source B N7:109
800<
OIL TEMP
B3:4
1
OIL OVERTEMP
O:1
111
1747-DCM-FULL
OVERTEMP
O:1
112
1747-DCM-FULL

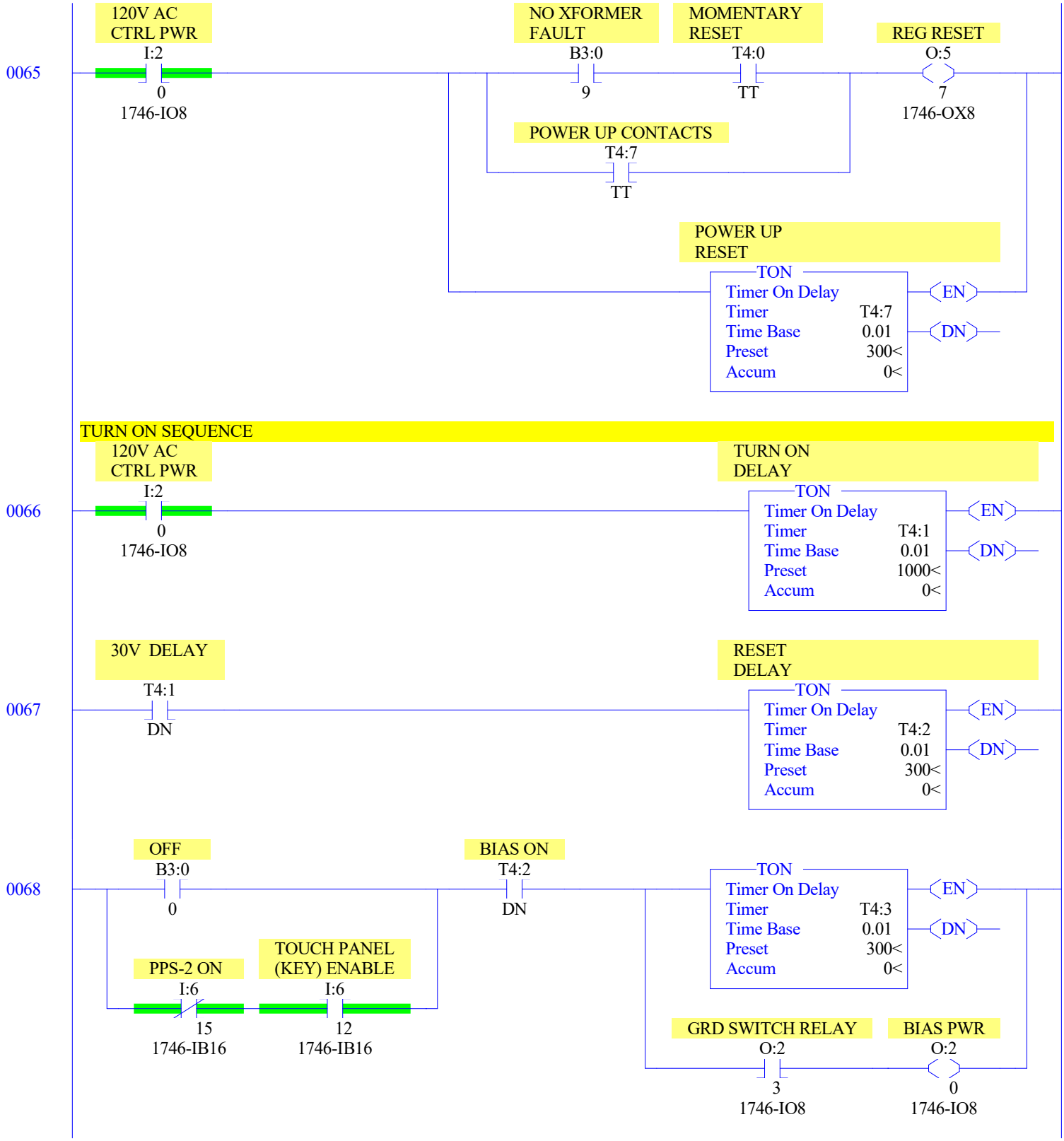

0044
RESET
B3:0
10
SCR OIL
TANK LEVEL
B3:4
2
CROWBAR
OIL LEVEL
I:6
10
1746-IB16
SCR OIL
LEVEL
I:6
11
1746-IB16
SCR OIL
TANK LEVEL
B3:4
2
0045
RESET
B3:0
10
LOW OIL
LATCH
B3:4
3
LOW OIL
LEVEL
I:7
7
1746-IV16
LOW OIL
B3:4
3
LOW OIL
O:1
110
1747-DCM-FULL
0046
RESET
B3:0
10
SUDDEN
PRESSURE
B3:4
4
XFRMER SUDDEN
PRESSURE
I:7
8
1746-IV16
SUDDEN
PRESSURE
B3:4
4
SUDDEN
PRESSURE
O:1
118
1747-DCM-FULL
AC INTERLOCKS LATCHING WITH RESET AND OVERRIDE
0047
RESET
B3:0
10
OIL PUMP
FLOW
B3:4
7
OIL PUMP
FLOW
I:7
9
1746-IV16
ON
LATCH
B3:0
4
OIL PUMP
FLOW
B3:4
7

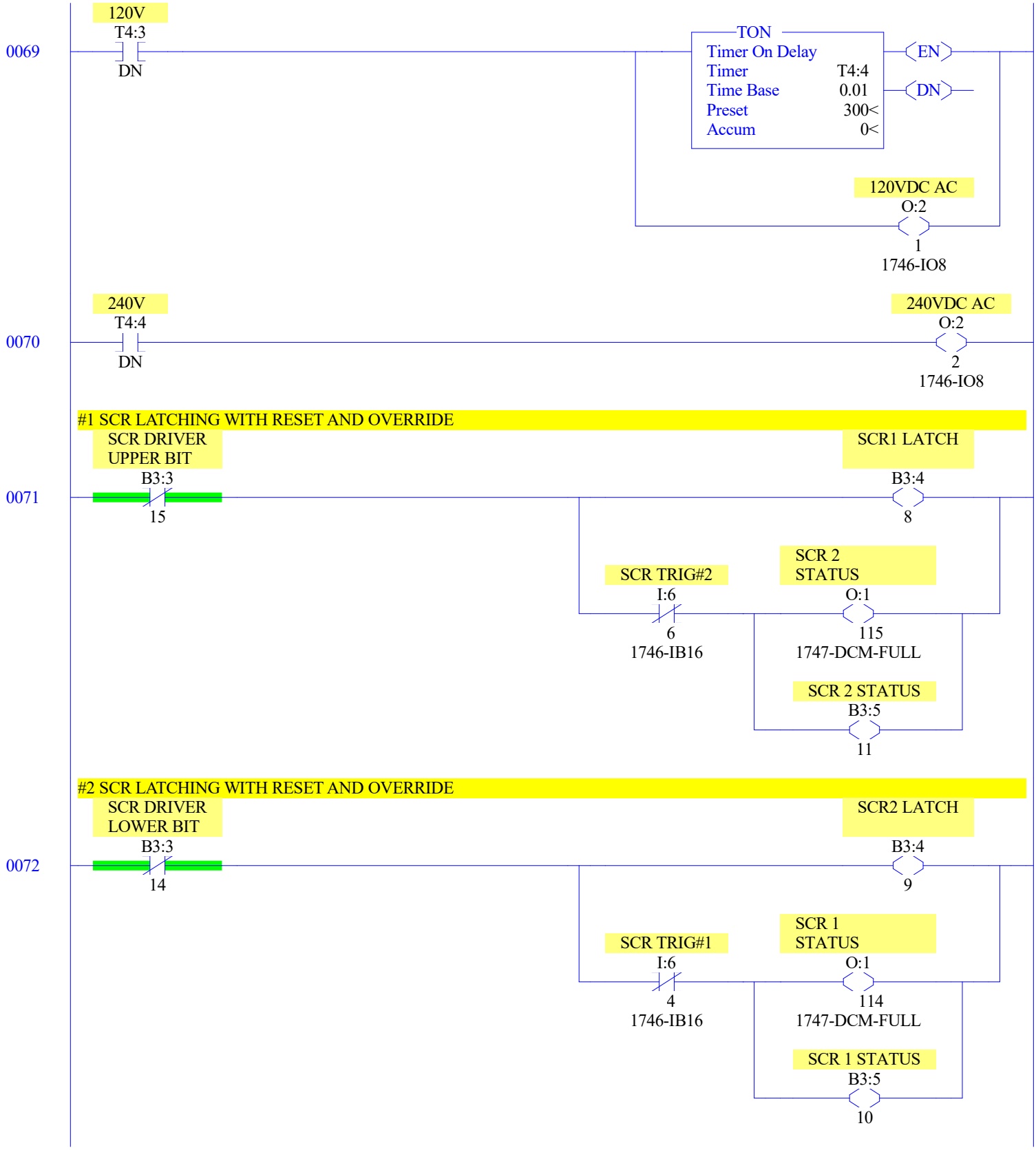

0048
RESET
B3:0
10
KLYSTRON CROWBAR LATCH
B3:2
6
SCR DISABLE FIBER DR
I:6
0
1746-IB16
CROWBAR ENABLE FIBER DR
I:6
1
1746-IB16
SCR ENABLE
O:5
0
1746-OX8
ARC TRIP KLYSTRON
I:6
3
1746-IB16
KLYSTRON CROWBAR LATCH
B3:2
6
0049
SCR DISABLE FIBER DR
I:6
0
1746-IB16
KLYSTRON SCR BIT
B3:2
5
0050
RESET
B3:0
10
H20_FLOW_SWITCH
B3:2
7
WATER_FLOW_SWITCH
I:7
10
1746-IV16
H20_FLOW_SWITCH
B3:2
7
DC POWER INTERLOCKS
0051
RESET
B3:0
10
PHASE LOSS
B3:2
3
PHASE LOSS
I:7
11
1746-IV16
ON LATCH
B3:0
4
PHASE LOSS
B3:2
3

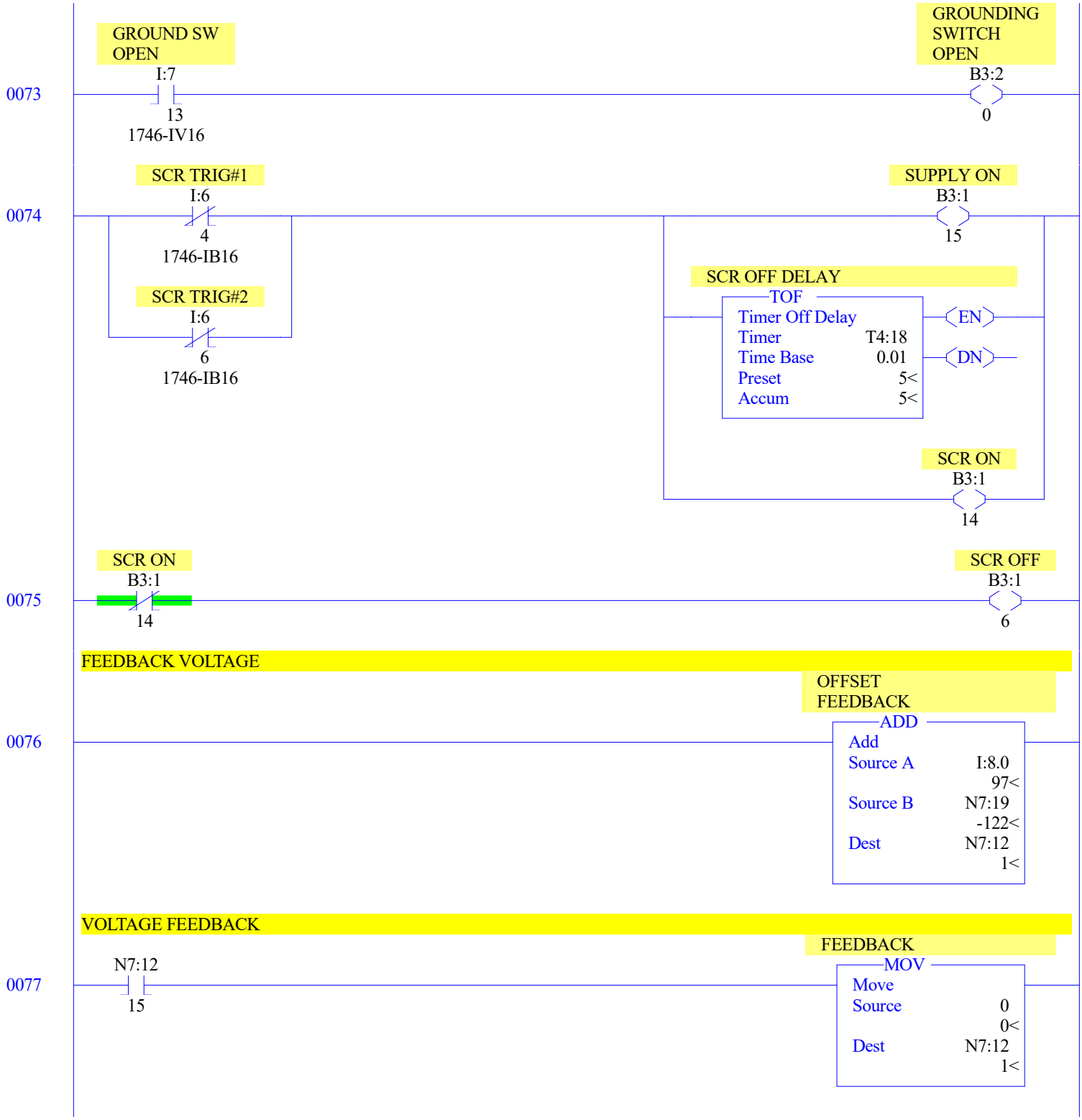

0052
RESET
B3:0
10
GRN TANK OIL LEVEL
B3:4
14
GRN TANK OIL LEVEL
I:6
8
1746-IB16
GRN TANK OIL LEVEL
B3:4
14
0053
RESET
B3:0
10
OVER CURRENT LATCH
B3:4
11
T4:2
TT
CONTACTOR OVER CURRENT
I:7
1
1746-IV16
REGULATOR CURRENT TRIP
I:7
15
1746-IV16
OVER CURRENT LATCH
B3:4
11
AC CURRENT TRIP
O:1
99
1747-DCM-FULL
0054
RESET
B3:0
10
OVER VOLT LATCH
B3:4
12
T4:2
TT
OVER VOLTAGE TRIP
I:7
14
1746-IV16
OVER VOLT LATCH
B3:4
12
OVER VOLT LATCH
O:1
113
1747-DCM-FULL
KLYSTRON ARC
0055
RESET
B3:0
10
KLYSTRON ARC TRIP
B3:4
13
ARC TRIP KLYSTRON
I:6
3
1746-IB16
CONTACTOR OPEN
I:7
2
1746-IV16
KLYSTRON ARC TRIP
B3:4
13
ARC TRIP
O:1
100
1747-DCM-FULL


KLYSTRON CROWBAR
0056
RESET
B3:0
10
KLYSTRON_CROWBAR
I:6
7
1746-IB16
_KLYSTRON_CROWBAR
B3:2
4
_KLYSTRON_CROWBAR
B3:2
4
DCM_BIT
O:1
123
1747-DCM-FULL
TRANSFORMER ARC
0057
RESET
B3:0
10
XFORMER ARC
I:6
5
1746-IB16
XFORMER ARC TRIP
B3:4
15
XFORMER ARC TRIP
B3:4
15
CONTACTOR OPEN
I:7
2
1746-IV16
O:1
122
1747-DCM-FULL
VACUUM CONTACTOR CURRENT FAULT LATCHING
0058
CONTACTOR LOCKOUT
I:7
0
1746-IV16
CONTACTOR OVER CURRENT
I:7
1
1746-IV16
OVER CURRENT TRIP LATCH
B3:4
5
CONTACTOR READY
I:7
3
1746-IV16
CONTACTOR OK
B3:2
2
CONTACTOR CLOSED
I:7
2
1746-IV16
DC POWER INTERLOCKS LATCHING LATCH RESET
0059
RESET
B3:0
10
AUX PWR ON
B3:1
8
DC POWER FAULT
B3:4
6
DC POWER LATCH
B3:4
6
OFF
B3:0
0


0060
PRESSURE LATCH
B3:4
0
OIL TEMP LATCH
B3:4
1
SCR OIL TANK LEVEL
B3:4
2
LOW OIL LATCH
B3:4
3
H20_FLOW_SWITCH
B3:2
7
XFORMER LATCHED OK
B3:2
1
0061
XFORMER PRESSURE ALARM
I:7
4
1746-IV16
XFORMER VACUUM
I:7
5
1746-IV16
XFORMER OVER TEMP
I:7
6
1746-IV16
LOW OIL LEVEL
I:7
7
1746-IV16
XFRMER SUDDEN PRESSURE
I:7
8
1746-IV16
CROWBAR OIL LEVEL
I:6
10
1746-IB16
SCR OIL LEVEL
I:6
11
1746-IB16
WATER_FLOW_SWITCH
I:7
10
1746-IV16
NO XFORMER FAULT
B3:0
9
0062
120V AC CTRL PWR
I:2
0
1746-IO8
AC AUX PWR ON
O:1
98
1747-DCM-FULL
0063
120V AC CTRL PWR
I:2
0
1746-IO8
240V PWR
O:2
2
1746-IO8
120V PWR
O:2
1
1746-IO8
AUX PWR ON DISPLAY
B3:1
8
AUX POWER
O:1
106
1747-DCM-FULL
KEY ENABLE
0064
KEY ON
B3:1
9
DISABLE DISPLAY
B3:1
2


0065
120V AC CTRL PWR
I:2
0
1746-IO8
NO XFORMER FAULT
B3:0
9
MOMENTARY RESET
T4:0
TT
REG RESET
O:5
7
1746-OX8
POWER UP CONTACTS
T4:7
TT
POWER UP RESET
TON
Timer On Delay
Timer T4:7
Time Base 0.01
Preset 300<
Accum 0<
EN
DN
TURN ON SEQUENCE
0066
120V AC CTRL PWR
I:2
0
1746-IO8
TURN ON DELAY
TON
Timer On Delay
Timer T4:1
Time Base 0.01
Preset 1000<
Accum 0<
EN
DN
0067
30V DELAY
T4:1
DN
RESET DELAY
TON
Timer On Delay
Timer T4:2
Time Base 0.01
Preset 300<
Accum 0<
EN
DN
0068
OFF
B3:0
0
BIAS ON
T4:2
DN
PPS-2 ON
I:6
15
1746-IB16
TOUCH PANEL (KEY) ENABLE
I:6
12
1746-IB16
TON
Timer On Delay
Timer T4:3
Time Base 0.01
Preset 300<
Accum 0<
EN
DN
GRD SWITCH RELAY
O:2
3
1746-IO8
BIAS PWR
O:2
0
1746-IO8


0069
120V
T4:3
DN
TON
Timer On Delay
Timer T4:4
Time Base 0.01
Preset 300<
Accum 0<
EN
DN
120VDC AC
O:2
1
1746-IO8
0070
240V
T4:4
DN
240VDC AC
O:2
2
1746-IO8
#1 SCR LATCHING WITH RESET AND OVERRIDE
0071
SCR DRIVER UPPER BIT
B3:3
15
SCR1 LATCH
B3:4
8
SCR TRIG#2
I:6
6
1746-IB16
SCR 2 STATUS
O:1
115
1747-DCM-FULL
SCR 2 STATUS
B3:5
11
#2 SCR LATCHING WITH RESET AND OVERRIDE
0072
SCR DRIVER LOWER BIT
B3:3
14
SCR2 LATCH
B3:4
9
SCR TRIG#1
I:6
4
1746-IB16
SCR 1 STATUS
O:1
114
1747-DCM-FULL
SCR 1 STATUS
B3:5
10


0073
GROUND SW OPEN
I:7
13
1746-IV16
GROUNDING SWITCH OPEN
B3:2
0
0074
SCR TRIG#1
I:6
4
1746-IB16
SCR TRIG#2
I:6
6
1746-IB16
SUPPLY ON
B3:1
15
SCR OFF DELAY
TOF
Timer Off Delay
Timer T4:18
Time Base 0.01
Preset 5<
Accum 5<
EN
DN
SCR ON
B3:1
14
0075
SCR ON
B3:1
14
SCR OFF
B3:1
6
FEEDBACK VOLTAGE
0076
OFFSET FEEDBACK
ADD
Add
Source A I:8.0
97<
Source B N7:19
-122<
Dest N7:12
1<
VOLTAGE FEEDBACK
0077
N7:12
15
FEEDBACK
MOV
Move
Source 0
0<
Dest N7:12
1<

0078

AC CURRENT MONITOR

ADD
Add
Source A I:9.0
136<
Source B N7:9
-117<
Dest N7:14
0<

AC CURRENT

0079

N7:14
15

AC CURRENT MONITOR

MOV
Move
Source 0
0<
Dest N7:14
0<

VOLTAGE MONITOR #1

0080

VOLTAGE MONITOR #1

MOV
Move
Source I:9.1
-1<
Dest N7:15
1<

VOLTAGE MONITOR #1

NEG
Negate
Source N7:15
1<
Dest N7:15
1<

OVERFLOW BIT

S:5
U
0

VOLTAGE MONITOR #2

0081

VOLTAGE MONITOR #2

MOV
Move
Source I:9.2
15<
Dest N7:16
0<

VOLTAGE MONITOR #2

NEG
Negate
Source N7:16
0<
Dest N7:16
0<

OVERFLOW BIT
S:5
U
0

DC CURRENT MONITOR

0082

DC CURRENT MONITOR

MUL
Multiply
Source A I:9.3
1<
Source B 1
1<
Dest N7:17
11<

DC CURRENT MONITOR

0083

DC CURRENT MONITOR
N7:17
15

CURRENT MONITOR

MOV
Move
Source 0
0<
Dest N7:17
11<

VOLTAGE MONITOR #1

0084

VOLTAGE #1

MUL
Multiply
Source A N7:12
1<
Source B N7:22
10075<
Dest N7:2
0<

OVERFLOW BIT
S:5
U
0

DDV
Double Divide
Source 32767
32767<
Dest N7:2
0<

0085

NEG DETECT
N7:2
15

MOV
Move
Source 0
0<
Dest N7:2
0<

DISPLAY VOLTAGE

0086

Time Base
S:4
2

B3:5
OSR
1

DISPLAY VOLTAGE

DIV
Divide
Source A N7:15
1<
Source B N7:47
320<
Dest N7:37
0<

PLOT VOLTAGE

0087

Time Base
S:4
2

B3:5
OSR
2

PLOT VOLTS

DIV
Divide
Source A N7:2
0<
Source B N7:44
43<
Dest N7:34
0<

PHASE ANGLE MONITOR

0088

PHASE MONITOR

MOV
Move
Source I:8.1
373<
Dest N7:13
21<

PHASE ANGL VOLTAGE

MUL
Multiply
Source A N7:13
21<
Source B N7:23
1000<
Dest N7:3
1<

OVERFLOW BIT

S:5
U
0

DDV
Double Divide
Source 32767
32767<
Dest N7:3
1<

PHASE ANGLE VOLTAGE

NEG DETECT

0089

N7:3
15

MOV
Move
Source 0
0<
Dest N7:3
1<

CURRENT MONITOR #1

0090

AC CURRENT MONITOR

MUL
Multiply
Source A N7:14
0<
Source B N7:24
4600<
Dest N7:4
0<

OVERFLOW BIT
S:5
U
0

AC CURRENT MONITOR

DDV
Double Divide
Source 32767
32767<
Dest N7:4
0<

STORE POWER

0091

AC CURRENT MONITOR
N7:4
15

AC CURRENT MONITOR

MOV
Move
Source 0
0<
Dest N7:4
0<

KLY VOLTAGE DISPLAY

0092

Time Base

S:4 3 — B3:5 OSR 8

AC CURRENT

MOV
Move
Source N7:4 0<
Dest O:1.1 0<

COP
Copy File
Source #I:3.0
Dest #N7:100
Length 8

REG VOLTS

MOV
Move
Source N7:10 0<
Dest O:1.2 0<

MONITOR VOLTAGE

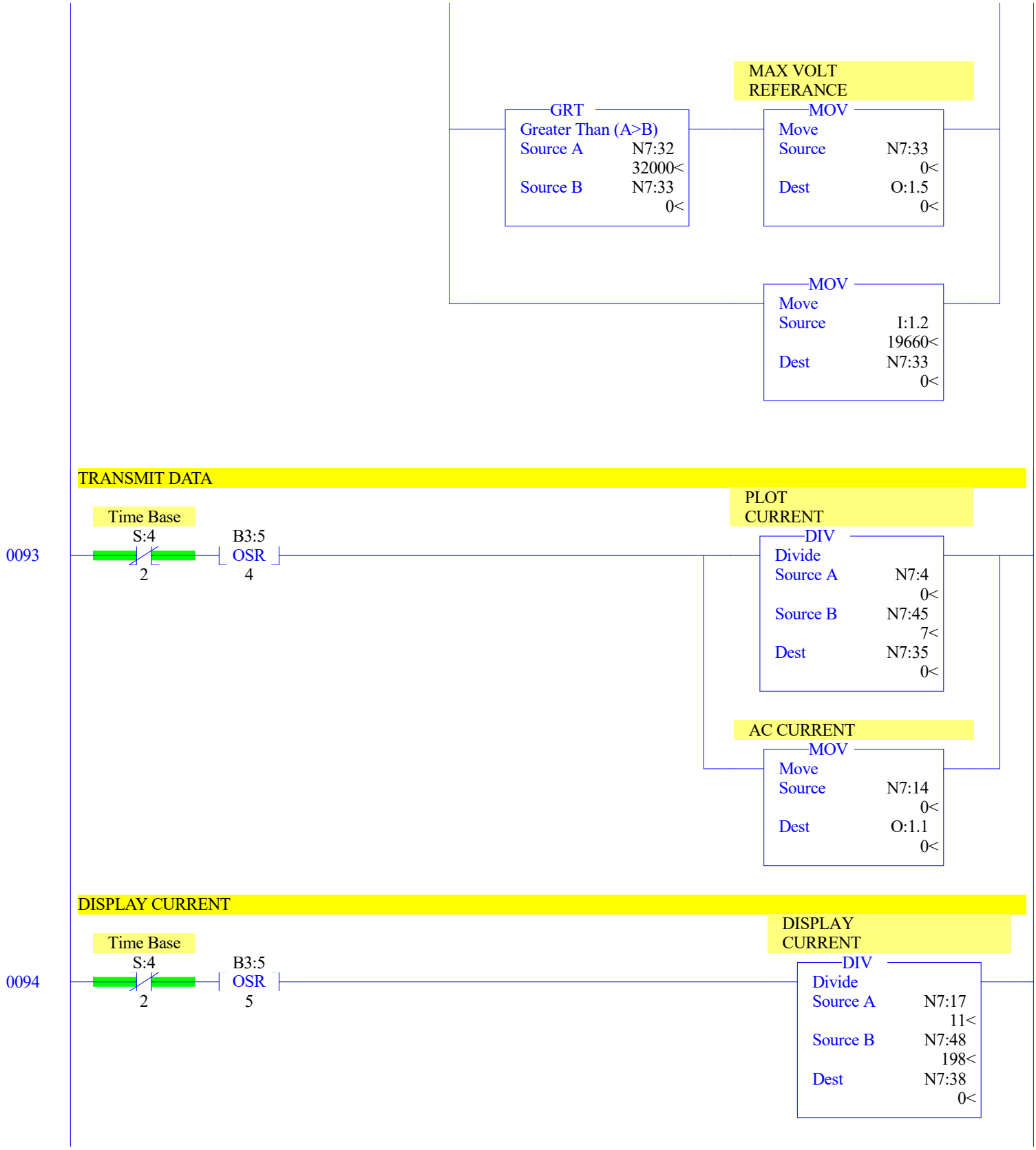
MOV
Move
Source N7:15 1<
Dest O:1.3 1<

DC CURRENT

MOV
Move
Source N7:17 11<
Dest O:1.4 11<

MAX VOLT REFERANCE

MOV
Move
Source N7:32 32000<
Dest O:1.5 0<


MAX VOLT REFERANCE
GRT
Greater Than (A>B)
Source A N7:32
32000<
Source B N7:33
0<
MOV
Move
Source N7:33
0<
Dest O:1.5
0<
MOV
Move
Source I:1.2
19660<
Dest N7:33
0<
TRANSMIT DATA
Time Base
S:4
2
B3:5
OSR
4
0093
PLOT CURRENT
DIV
Divide
Source A N7:4
0<
Source B N7:45
7<
Dest N7:35
0<
AC CURRENT
MOV
Move
Source N7:14
0<
Dest O:1.1
0<
DISPLAY CURRENT
Time Base
S:4
2
B3:5
OSR
5
0094
DISPLAY CURRENT
DIV
Divide
Source A N7:17
11<
Source B N7:48
198<
Dest N7:38
0<

0095

VOLTAGE X CURRENT

MUL
Multiply
Source A N7:15
1<
Source B N7:17
11<
Dest N7:18
0<

OVERFLOW BIT
S:5
U
0

VOLTAGE X CURRENT

DDV
Double Divide
Source 32767
32767<
Dest N7:18
0<

POWER METER

0096

Time Base
S:4
2

B3:5
OSR
6

PLOT METER

DIV
Divide
Source A N7:18
0<
Source B N7:49
162<
Dest N7:39
0<

DISPLAY POWER

0097

Time Base
S:4
2

B3:5
OSR
7

DISPLAY POWER

DIV
Divide
Source A N7:18
0<
Source B N7:46
260<
Dest N7:36
0<

VOLTAGE MONITOR #2

0098

VOLTAGE MONITOR #2

MUL
Multiply
Source A N7:15
1<
Source B N7:25
10000<
Dest N7:5
0<

OVERFLOW BIT
S:5
U
0

DDV
Double Divide
Source 32767
32767<
Dest N7:5
0<

0099

N7:5
15

DC CURRENT

MOV
Move
Source 0
0<
Dest N7:5
0<

AC VOLTAGE

0100

AC VOLTAGE

MUL
Multiply
Source A N7:16
0<
Source B N7:26
10000<
Dest N7:6
0<

OVERFLOW BIT
S:5
U
0

DDV
Double Divide
Source 32767
32767<
Dest N7:6
0<

0101

N7:6
15

MOV
Move
Source 0
0<
Dest N7:6
0<

CURRENT #2

0102

DC
CURRENT #2

MUL
Multiply
Source A N7:17
11<
Source B N7:27
5000<
Dest N7:7
0<

OVERFLOW
BIT
S:5
U
0

DDV
Double Divide
Source 32767
32767<
Dest N7:7
0<

NEG DETECT

0103

N7:7
15

MOV
Move
Source 0
0<
Dest N7:7
0<

0104

REGULATOR ON B3:0 2 — RATE S:4 2 — SINGLE SHOT B3:5 OSR 0

VOLTAGE REFERANCE
MOV
Move
Source I:1.1 0<
Dest N7:30 100<

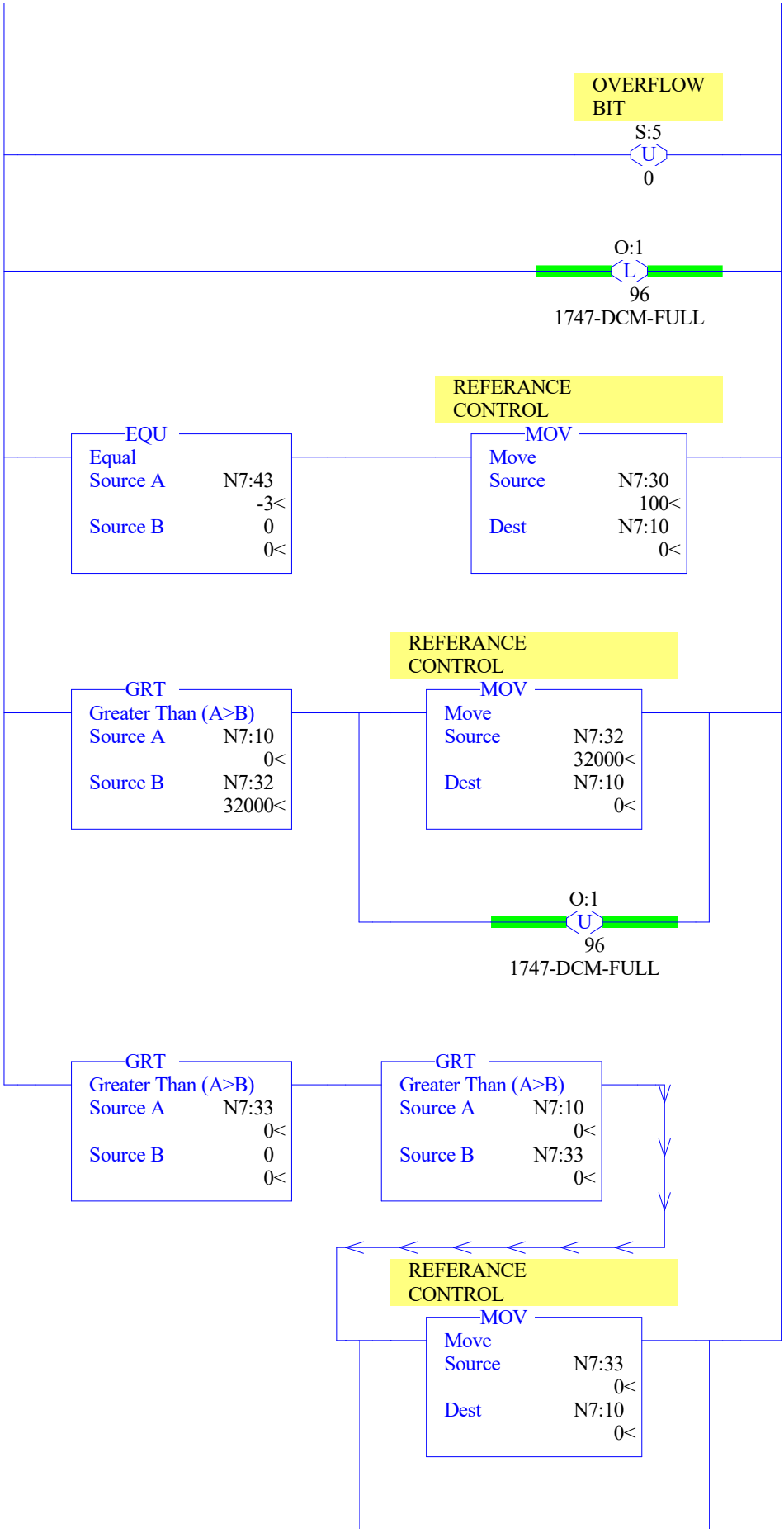
VOLTAGE REFERANCE
GRT
Greater Than (A>B)
Source A N7:31 100<
Source B N7:30 100<

MOV
Move
Source N7:31 100<
Dest N7:30 100<

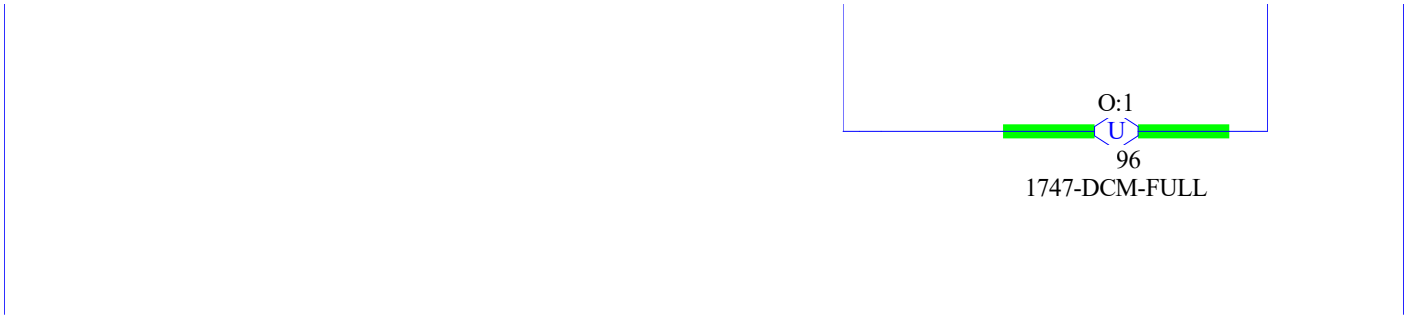
SUB
Subtract
Source A N7:30 100<
Source B N7:10 0<
Dest N7:43 -3<

DIV
Divide
Source A N7:43 -3<
Source B 10 10<
Dest N7:43 -3<

ADD
Add
Source A N7:10 0<
Source B N7:43 -3<
Dest N7:10 0<


OVERFLOW BIT
S:5
U
0
O:1
L
96
1747-DCM-FULL
REFERANCE CONTROL
EQU
Equal
Source A N7:43
-3<
Source B 0
0<
MOV
Move
Source N7:30
100<
Dest N7:10
0<
REFERANCE CONTROL
GRT
Greater Than (A>B)
Source A N7:10
0<
Source B N7:32
32000<
MOV
Move
Source N7:32
32000<
Dest N7:10
0<
O:1
U
96
1747-DCM-FULL
GRT
Greater Than (A>B)
Source A N7:33
0<
Source B 0
0<
GRT
Greater Than (A>B)
Source A N7:10
0<
Source B N7:33
0<
REFERANCE CONTROL
MOV
Move
Source N7:33
0<
Dest N7:10
0<

LAD 2 - --- Total Rungs in File = 120



0105

Time Base
S:4
5

B3:5
OSR
9

MAX
VOLTAGE

MOV
Move
Source N7:71
32000<
Dest N7:32
32000<

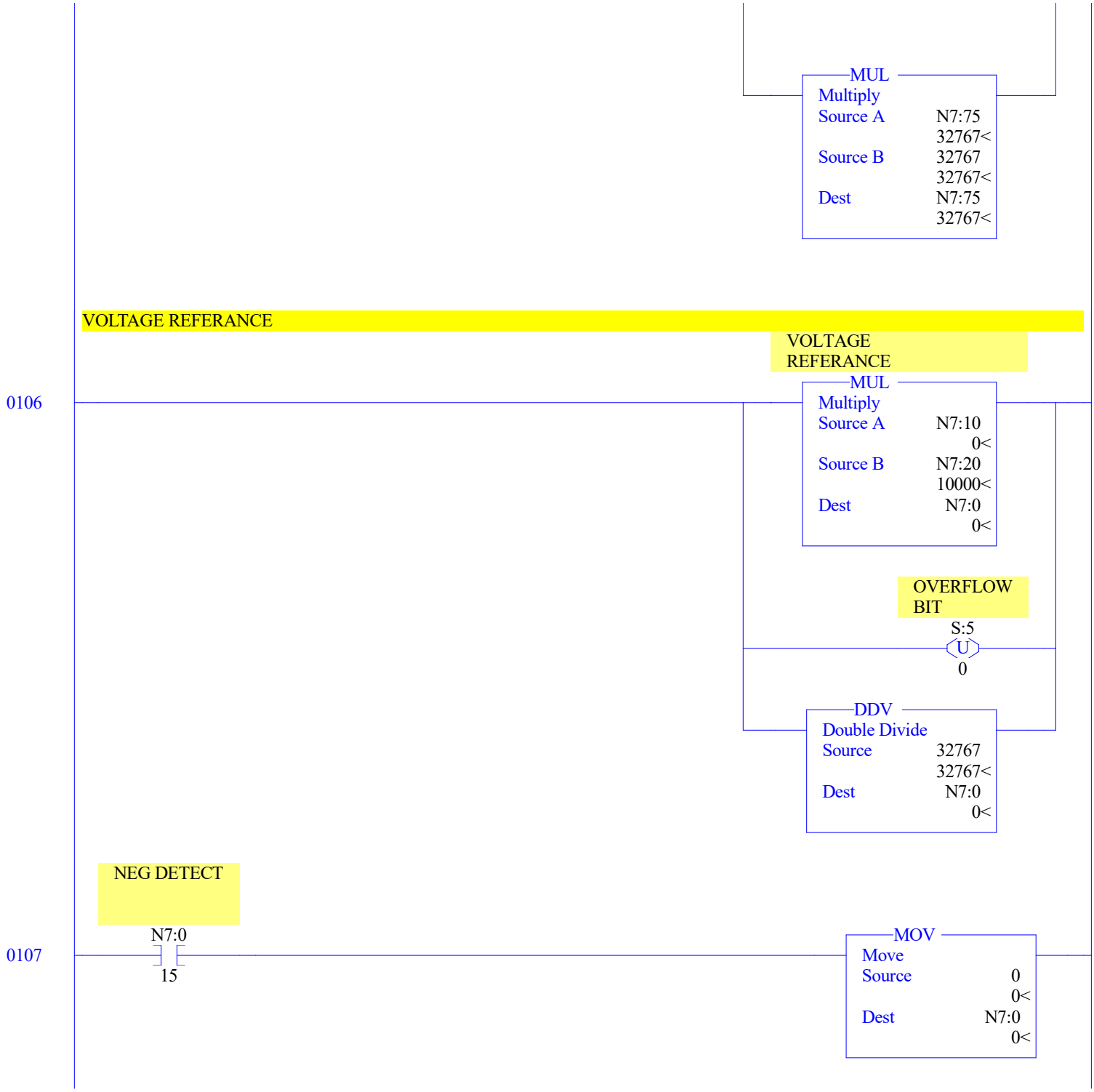
MOV
Move
Source N7:72
18000<
Dest N7:42
18000<

MOV
Move
Source N7:73
100<
Dest N7:31
100<

DIV
Divide
Source A N7:71
32000<
Source B N7:10
0<
Dest N7:74
32767<

MUL
Multiply
Source A N7:74
32767<
Source B 32767
32767<
Dest N7:74
32767<

DIV
Divide
Source A N7:73
100<
Source B N7:10
0<
Dest N7:75
32767<


MUL
Multiply
Source A N7:75
32767<
Source B 32767
32767<
Dest N7:75
32767<
VOLTAGE REFERANCE
VOLTAGE REFERANCE
0106
MUL
Multiply
Source A N7:10
0<
Source B N7:20
10000<
Dest N7:0
0<
OVERFLOW BIT
S:5
U
0
DDV
Double Divide
Source 32767
32767<
Dest N7:0
0<
NEG DETECT
N7:0
15
0107
MOV
Move
Source 0
0<
Dest N7:0
0<

PHASE ANGLE CONTROL LIMIT

0108

SAMPLING
B3:5
0

PHASE ANGLE CONTROL

MOV
Move
Source N7:10
0<
Dest N7:11
0<

PHASE ANGLE MULTIPLIER

MUL
Multiply
Source A N7:11
0<
Source B N7:40
12000<
Dest N7:11
0<

OVERFLOW BIT
S:5
U
0

DDV
Double Divide
Source N7:11
0<
Dest N7:11
0<

ADD OFFSET

ADD
Add
Source A N7:11
0<
Source B N7:41
6000<
Dest N7:11
0<

OVERFLOW BIT
S:5
U
0

PHASE ANGLE INPUT

0109

GRT
Greater Than (A>B)
Source A N7:11
0<
Source B N7:42
18000<

PHASE ANGLE CONTROL

MOV
Move
Source N7:42
18000<
Dest N7:11
0<

PHASE ANGLE DISPLAY

0110

PHASE ANGLE

MUL
Multiply
Source A N7:11
0<
Source B N7:21
1000<
Dest N7:1
0<

OVERFLOW BIT
S:5
U
0

DDV
Double Divide
Source 32767
32767<
Dest N7:1
0<

NEG DETECT

0111

N7:1
15

MOV
Move
Source 0
0<
Dest N7:1
0<

0112

MOV
Move
Source N7:10
0<
Dest O:8.0
0<

0113

MOV
Move
Source N7:11
0<
Dest O:8.1
15425<

DISPLAY POWER

0114

DISPLAY POWER

DIV
Divide
Source A N7:18 0<
Source B N7:29 6<
Dest N7:8 0<

OVERFLOW BIT
S:5
U
0

RESET CONTROLS ON POWER UP AND RESET BIT RESETS. TEMPERATURE MONITORS RESET

0115

RESET BIT
B3:0
11

CONTROL RESET
I:1
80
1747-DCM-FULL

POWER UP CONTACTS
T4:7
TT

First Pass
S:1
15

MOMENTARY RESET DELAY

TON
Timer On Delay
Timer T4:0
Time Base 0.01
Preset 100<
Accum 0<
EN
DN

B3:12
OSR
8

COP
Copy File
Source #N7:90
Dest #O:3.0
Length 4

ENERPRO PHASE LOSS - CONTACTOR CLOSED

0116

CONTACTOR CLOSED
B3:1
11

PHASE LOSS
I:7
11
1746-IV16

_ENERPRO_PHASE_LOSS
B3:5
L
3

0117

1280 mS Time Base
S:4
6

B3:17
OSR
0

COPY I/0 TO B3 SBR

JSR
Jump To Subroutine
SBR File Number U:3

0118

2560 mS Time Base
S:4
7

B3:17
OSR
1

SCALE VALUES SBR

JSR
Jump To Subroutine
SBR File Number U:4

0119 ——— END

LAD 3 - COPY - COPY O0 AND I1 WORDS TO B3 FOR QUICKPANEL DISPLAY --- Total Rungs in File = 6

COPY I/O DATA FILE WORDS TO B3

INPUT MODULE SLOT 2

0000

COP
Copy File
Source #I:2.0
Dest #B3:12
Length 1

INPUT MODULE SLOT 6

0001

COP
Copy File
Source #I:6.0
Dest #B3:13
Length 1

INPUT MODULE SLOT 7

0002

COP
Copy File
Source #I:7.0
Dest #B3:14
Length 1

OUTPUT MODULE SLOT 2

0003

COP
Copy File
Source #O:2.0
Dest #B3:15
Length 1

OUTPUT MODULE SLOT 5

0004

COP
Copy File
Source #O:5.0
Dest #B3:16
Length 1

0005

END

SCALE THERMOCOUPLE VALUES FOR QUICKPANEL DISPLAY

TC1 - SCR TOP OIL

0000

SCP
Scale w/Parameters

Input	N7:100
	0<
Input Min.	0
	0<
Input Max.	999
	999<
Scaled Min.	0
	0<
Scaled Max.	9999
	9999<
Output	N7:110
	0<

TC2 - SCR BOTTOM OIL

0001

SCP
Scale w/Parameters

Input	N7:101
	0<
Input Min.	0
	0<
Input Max.	999
	999<
Scaled Min.	0
	0<
Scaled Max.	9999
	9999<
Output	N7:111
	0<

TC3 - CROWBAR OIL

0002

SCP
Scale w/Parameters

Input	N7:102
	192<
Input Min.	0
	0<
Input Max.	999
	999<
Scaled Min.	0
	0<
Scaled Max.	9999
	9999<
Output	N7:112
	0<

LAD 4 - SCALE - SCALE THERMOCOUPLE VALUES FOR QUICKPANEL DISPLAY --- Total Rungs in File = 5

TC4 - CONTROL CABINET AIR TEMPERATURE

0003

SCP
Scale w/Parameters

Input	N7:103
	287<
Input Min.	0
	0<
Input Max.	999
	999<
Scaled Min.	0
	0<
Scaled Max.	9999
	9999<
Output	N7:113
	0<

0004 END